Linear Algebra and Optimization

The answers here are for the problems in [Gallier and Quaintance, 2020]. They reflect my effort to answer them and in no way should be construed as the approved solutions for this book.

Chapter 2

Problem 2.1

1. To prove that $(H, \cdot)$ is a group we must show that the following properties hold:

(1) **Closure**: if $A, B \in H$ then $A \cdot B \in H$. To show this note that any upper triangular matrix A has the property

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a & b \\ d & 1 & c \\ e & f & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

where $\odot$ denotes the **Hadamard** product of two matrices and the first matrix is non-singular. In index notation we write this as

$$A_{ij} = a_{ij} \mathbb{1}_{j \geq i}$$

where $\mathbb{1}_{\text{statement}} = 1$ if the statement is true and zero otherwise. Then

$$\begin{aligned} (A \cdot B)_{ij} &= \sum_{k=1}^{n} A_{ik} B_{kj} = \sum_{k=1}^{n} a_{ik} b_{kj} \mathbb{1}_{k \geq i} \mathbb{1}_{j \geq k} \\ &= \sum_{k=1}^{n} a_{ik} b_{kj} \mathbb{1}_{j \geq i} = (a \cdot b)_{ij} \mathbb{1}_{j \geq i} \end{aligned}$$

and so, $A \cdot B \in H$.

(2) Since H is a subspace of the vector space of 3×3 matrices it inherits associativity and the identity element.

(3) The inverse A^{-1} of $A \in H$ must have the property

$$A \cdot A^{-1} = A^{-1} \cdot A = I, \quad A^{-1} \in H.$$

We note that, since we can write any uppper-triangular matrix as the Hadamard product of a standard matrix and an upper-triangular matrix of ones, then

$$(A \cdot A^{-1})_{ij} = \sum_{k=1}^{n} a_{ik} a_{kj}^{-1} \mathbb{1}_{k \geq i} \mathbb{1}_{j \geq k} = \sum_{k=1}^{n} \delta_{ij} \mathbb{1}_{k \geq i} \mathbb{1}_{j \geq k} = \delta_{ij}$$

where $\delta_{ij} = 1$ if $i = j$ and zero otherwise. Therefore the inverse exists and is an element of H. The following explicit form of the inverse

$$\begin{bmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{bmatrix}$$

satisfies the diagonal and lower triangular conditions. For the upper triangular elements we must have

$$\begin{aligned} a' + a &= 0 \\ b' + ac' + b &= 0 \\ c' + c &= 0 \end{aligned}$$

and so

$$A^{-1} = \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}.$$

If $(H, \cdot)$ is abelian then $A \cdot B = B \cdot A$ or

$$(A \cdot B)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \mathbb{1}_{k \geq i} \mathbb{1}_{j \geq k} = \mathbb{1}_{j \geq i} \sum_{k=1}^{n} a_{ik} b_{kj} = \mathbb{1}_{j \geq i} (a \cdot b)_{ij},$$
$$(B \cdot A)_{ij} = \sum_{k=1}^{n} b_{ik} a_{kj} \mathbb{1}_{k \geq i} \mathbb{1}_{j \geq k} = \mathbb{1}_{j \geq i} \sum_{k=1}^{n} b_{ik} a_{kj} = \mathbb{1}_{j \geq i} (b \cdot a).$$

Since $a \cdot b \neq b \cdot a$ we conclude that H is not abelian.

2. If $a \in G_1$ then from the definition of e_1

$$\phi(ae_1) = \phi(a) \in G_2$$

and since ϕ is a homomorphism

$$\phi(ae_1) = \phi(a)\phi(e_1).$$

Hence

$$\phi(a)\phi(e_1) = \phi(a)$$

and we conclude that $\phi(e_1) = e_2$.

Since $aa^{-1} = e_1$

$$\phi(aa^{-1}) = \phi(a)\phi(a^{-1}) = \phi(e_1).$$

We have already shown that $\phi(e_1) = e_2$. Therefore $\phi(a^{-1}) = \phi(a)^{-1}$.

3. For any $A \in H$, elements $a, b, c \in \mathbb{R}$. Since if $b \in \mathbb{R}$, e^{ib} generates all elements of S^1 we conclude that ϕ is surjective.

To prove that $(G, \cdot)$ first note that closure is a consequence of the closure of $\mathbb{R}$ w.r.t. addition and multiplication and the fact that if $u_1, u_2 \in S^1$ then, since $e^{ix_1y_2} \in S_1$, $e^{ix_1y_2}u_1u_2 \in S_1$ (closure of S^1 under multiplication). To prove associativity write

$$\begin{aligned}
&((x_1, y_1, u_1) \cdot (x_2, y_2, u_2)) \cdot (x_3, y_3, u_3) = \\
&\quad = (x_1 + x_2, y_1 + y_2, e^{ix_1y_2}u_1u_2) \cdot (x_3, y_3, u_3) = \\
&\quad = (x_1 + x_2 + x_3, y_1 + y_2 + y_3, e^{i(x_1+x_2)y_3}e^{ix_1y_2}u_1u_2u_3) = \\
&\quad = (x_1 + (x_2 + x_3), y_1 + (y_2 + y_3), e^{ix_1(y_2+y_3)}u_1(e^{ix_2y_3}u_2u_3)) = \\
&\quad = (x_1, y_1, u_1) \cdot (x_2 + x_3, y_2 + y_3, e^{ix_2y_3}u_2u_3) = \\
&\quad = (x_1, y_1, u_1) \cdot ((x_2, y_2, u_2) \cdot (x_3, y_3, u_3)).
\end{aligned}$$

The identity element of G, $(0, 0, 1)$ borrows the identity elements of$(\mathbb{R}, +)$ and $(S^1, \cdot)$. The inverse of (x, y, u) must have the property

$$(x, y, u) \cdot (x', y', u') = (x + x', y + y', e^{ixy'}uu') = (0, 0, 1)$$

and so $x' = -x,\ y' = -y,\ u' = e^{ixy}u^{-1}$. ϕ is a homomorphism if for $A, B \in H$

$$\phi(A \cdot B) = \phi(A) \cdot \phi(B).$$

Since

$$\begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a+d & af+b+e \\ 0 & 1 & c+f \\ 0 & 0 & 1 \end{bmatrix},$$

we have

$$\begin{aligned}\phi(A\cdot B) &= (a+d, c+f, e^{i(af+b+e)})\\ &= (a,c,e^{ib})\cdot(d,f,e^{ie})\\ &= \phi(A)\cdot\phi(B).\end{aligned}$$

4. We are looking for a set of matrices $A \in H$ such that

$$\phi(A) = (0,0,1)$$

the identity element in G. From the definition of ϕ we must have

$$a = 0, c = 0, b = 2\pi n$$

where $n \in \mathbb{Z}$. For any $A \in \ker\phi$ we can write $A = I + 2\pi n E_{13}$. Closure follows since if $A, B \in \ker\phi$

$$\begin{aligned}A\cdot B &= (I+2\pi E_{13})\cdot(I+2\pi m E_{13})\\ &= I + 2\pi(n+m)E_{13} + (2\pi)^2 nmE_{13}^2\\ &= I + 2\pi(n+m)E_{13} \in \ker\phi.\end{aligned}$$

Associativity is inherited; the identity element is I $(n=0)$ while the inverse of $A = I + 2\pi n E_{13}$ is $A^{-1} = I - 2\pi n E_{13}$. Hence $\ker\phi$ is a subgroup of H.

Problem 2.2

To check that $m\mathbb{Z} = \{mk | k \in \mathbb{Z}\}$ is an abelian subgroup of $\mathbb{Z}$ first note that $mk_1 + mk_2 = m(k_1+k_2)$ (closure). Addition is commutative/associative in $\mathbb{Z}$ and $m\mathbb{Z}$ inherits these properties. It also inherits the identity element $(k=0)$ and the inverse $(k^{-1} = -k)$.
1. $\phi(mk) := k$; this is an homomorphism since

$$\phi(mk_1 + mk_2) = \phi(m(k_1+k_2)) = k_1 + k_2 = \phi(mk_1) + \phi(mk_2).$$

Since $\ker\phi = \{0\}$ it is also an isomorphism.
2. If $i(mk) := mk$ then

$$i(mk_1 + mk_2) = i(m(k_1+k_2)) = m(k_1+k_2) = mk_1 + mk_2 = i(mk_1) + i(mk_2)$$

hence the inclusion map is a group homomorphism. If $p : \mathbb{Z} \to m\mathbb{Z}$ is a homomorphism then

$$p(l_1 + l_2) = p(l_1) + p(l_2) = mk_1 + mk_2$$

where $l_1, l_2, k_1, k_2 \in \mathbb{Z}$ (for example $k = \lfloor l/m \rfloor$). If $p \circ i = \mathrm{id}$ then p is the left inverse of i. By Proposition 2.16, i must be an isomorphism. However, unless $m = 1$, i is not an isomorphism since it is not surjective.

Problem 2.3

As defined E is an abelian group w.r.t. $+$. From the definition of $\cdot$ we have

$$\begin{aligned}\lambda\cdot((x_1,y_1)+(x_2,y_2)) &= \lambda\cdot(x_1+x_2, y_1+y_2) = (\lambda(x_1+x_2), y_1+y_2) = (\lambda x_1 + \lambda x_2, y_1+y_2)\\ &= (\lambda x_1, y_1) + (\lambda x_2, y_2) = \lambda\cdot(x_1,y_1) + \lambda\cdot(x_2,y_2)\end{aligned}$$

so axiom (V1) holds.

$$\begin{aligned}(\lambda+\mu)\cdot(x,y) &= ((\lambda+\mu)x, y) = (\lambda x + \mu x, y) = (\lambda x, y) + (\mu x, 0)\\ &= \lambda\cdot(x,y) + \mu\cdot(x,0) \neq \lambda\cdot(x,y) + \mu\cdot(x,y)\end{aligned}$$

so axiom (V2) fails.

$$(\lambda\mu)\cdot(x,y) = ((\lambda\mu)x, y) = (\lambda(\mu x), y) = \lambda\cdot(\mu x, y) = \lambda\cdot(\mu\cdot(x,y))$$

so axiom (V3) holds.

$$1\cdot(x,y) = (1 * x, y) = (x,y)$$

so axiom (V4) holds.

Problem 2.4

1. Below we (mostly) omit steps that use the properties of a field.

$$\begin{aligned}
\alpha\cdot 0 &= \alpha\cdot(u-u), && \text{any vector } u \text{ has an additive inverse}\\
&= \alpha\cdot u - \alpha\cdot u, && \text{(V1)}\\
&= 0, && \text{the additive inverse of } \alpha\cdot u \text{ is } -\alpha\cdot u
\end{aligned}$$

$$\begin{aligned}
0\cdot v &= (\alpha-\alpha)\cdot v, && K \text{ is a field}\\
&= \alpha\cdot v - \alpha\cdot v, && \text{(V2)}\\
&= 0, && \text{the additive inverse of } \alpha\cdot v \text{ is } -\alpha\cdot v
\end{aligned}$$

$$\begin{aligned}
\alpha\cdot(-v) &= \alpha\cdot(-1\cdot v)\\
&= (\alpha * -1)\cdot v, && \text{(V3)}\\
&= -(\alpha\cdot v)
\end{aligned}$$

$$\begin{aligned}
(-\alpha)\cdot v &= (-1 * \alpha)\cdot v\\
&= -1\cdot(\alpha\cdot v), && \text{(V3)}\\
&= -(\alpha\cdot v), && \text{the additive inverse of } \alpha\cdot v \text{ is } -1\cdot(\alpha\cdot v)
\end{aligned}$$

2. Axiom (V2) states that

$$\alpha\cdot(u+v) = \alpha\cdot u + \alpha\cdot v$$

where u, v are vectors and α is a scalar. Then

$$\begin{aligned}
\alpha\cdot x &= \alpha\cdot(x_1e_1 + x_2e_2 + \cdots + x_ne_n)\\
&= \alpha\cdot(x_1e_1 + (x_2e_2 + \cdots + x_ne_n))\\
&= \alpha\cdot(x_1e_1) + \alpha\cdot(x_2e_2 + \cdots + x_ne_n)\\
&\vdots\\
&= \alpha\cdot(x_1e_1) + \cdots + \alpha\cdot(x_{n-2}e_{n-2}) + \alpha\cdot(x_{n-1}e_{n-1} + x_ne_n)\\
&= \alpha\cdot(x_1e_1) + \cdots + \alpha\cdot(x_ne_n).
\end{aligned}$$

Since for any $i = 1, 2, \ldots, n$,

$$\alpha\cdot(x_ie_i) = \alpha\cdot(x_i\cdot e_i) = (\alpha * x_i)\cdot e_i = (x_i * \alpha)\cdot e_i = x_i\cdot(\alpha\cdot e_i)$$

given $\{\alpha\cdot e_i\}_{i=1}^n$ we can determine the result of scalar multiplication on any vector by its action on the basis vectors.

3. Lets work backwards. We want to define scalar multiplication so that

$$1\cdot u \neq u;$$

if we denote $1 \cdot u = \nu(u)$ then if axiom (V3) holds

$$(1 * 1) \cdot u = 1 \cdot (1 \cdot u) = 1 \cdot \nu(u) = \nu^2(u).$$

Since $1 * 1 = 1$ we conclude that $\nu(u) = \nu^2(u)$ and so, in this setting, scalar multiplication must be a projection. One example is

$$\alpha \cdot x = (\alpha x_1, \ldots, \alpha x_{n-1}, 0)$$

which for $\alpha = 1$ gives the desired result. All other axioms follow; for example

$$\begin{aligned}\alpha \cdot (x + y) &= (\alpha(x_1 + y_1), \ldots, \alpha(x_{n-1} + y_{n-1}), 0) \\ &= (\alpha x_1, \alpha x_{n-1}, 0) + (\alpha y_1, \ldots, \alpha y_{n-1}, 0) \\ &= \alpha \cdot x + \alpha \cdot y.\end{aligned}$$

4. If $n \in \mathbb{N}$ then according to (V2)

$$\begin{aligned}n \cdot x &= (1 + \cdots + 1) \cdot x \\ &= 1 \cdot x + \cdots + 1 \cdot x \\ &= (y_1, \ldots, y_n) + \cdots + (y_1, \ldots, y_n) \\ &= (ny_1, \ldots, ny_n) = n(y_1, \ldots, y_n) \\ &= n(1 \cdot x).\end{aligned}$$

Next use (V3):

$$1 \cdot x = \left(\frac{1}{n} * n\right) \cdot x = n \cdot \left(\frac{1}{n} \cdot x\right) = (1 + \cdots + 1) \cdot \left(\frac{1}{n} \cdot x\right).$$

If $\frac{1}{n} \cdot x = (y_1', \ldots, y_n')$ then since $1 \cdot x = (y_1, \ldots, y_n)$ we must have

$$(y_1, \ldots, y_n) = (ny_1', \ldots, ny_n').$$

This is possible only if $y_i' = y_i/n$ for $i = 1, \ldots, n$ and so we find that

$$\frac{1}{n} \cdot x = \frac{1}{n}(1 \cdot x).$$

A rational number $r = m/n$ where $m, n \in \mathbb{N}$; then

$$\begin{aligned}\frac{m}{n} \cdot x &= \left(m * \frac{1}{n}\right) \cdot x = m \cdot \left(\frac{1}{n} \cdot x\right) = m \cdot \left(\frac{1}{n}(1 \cdot x)\right) \\ &= \frac{1}{n}(m \cdot (1 \cdot x)) = \frac{1}{n}(1 \cdot (m \cdot x)) = \frac{1}{n}(1 \cdot m(1 \cdot x)) \\ &= \frac{m}{n}(1 \cdot (1 \cdot x)) = r((1 * 1) \cdot x) = r(1 \cdot x).\end{aligned}$$

Any vector x has the following expansion given a basis $\{e_1, \ldots, e_n\}$:

$$x = x_1 e_1 + \cdots + x_n e_n.$$

Given $1 \cdot x = y$ and

$$y = y_1 e_1 + \cdots + y_n e_n$$

we have

$$1 \cdot x = x_1(1 \cdot e_1) + \cdots + x_n(1 \cdot e_n).$$

Define $1 \cdot e_i = \sum_{j=1}^n \nu_{ji} e_j$ where ν_{ji} are scalars. Then

$$y_i = \sum_{j=1}^n \nu_{ji} x_i.$$

So, given any vector x, and the coefficients ν that determine the action of 1 on a basis of $\mathbb{R}^n$ we can determine $1 \cdot x$. Then since $r \cdot x = ry$ we can determine the action of any scalar multiplication.

Problem 2.5

Linear independence requires that

$$a\begin{pmatrix}2\\1\\-3\end{pmatrix}+b\begin{pmatrix}3\\2\\-5\end{pmatrix}+c\begin{pmatrix}1\\-1\\1\end{pmatrix}=0\Rightarrow a=b=c=0.$$

Starting from

$$\begin{aligned}2a+3b+c&=0\\-3a-5b+c&=0\end{aligned}$$

we obtain

$$5a-8b=0.$$

Then from

$$\begin{aligned}2a+3b+c&=0\\a+2b-c&=0\end{aligned}$$

we have

$$3a+5b=0.$$

The only solution to this system of equations is $a=b=0$ since

$$3(5a-8b)-5(3a+5b)=-b=0$$

and then $5a=0$. Substitute $a=b=0$ in any of the three equations to obtain $c=0$. The basis expansion of $x=(6,2,-7)$ in terms of these three independent vectors requires that

$$a\begin{pmatrix}2\\1\\-3\end{pmatrix}+b\begin{pmatrix}3\\2\\-5\end{pmatrix}+c\begin{pmatrix}1\\-1\\1\end{pmatrix}=\begin{pmatrix}6\\2\\-7\end{pmatrix}.$$

It is easy to show that $a=b=c=1$.

Problem 2.6

We want to show that

$$\begin{pmatrix}1\\3\\-1\\0\end{pmatrix}=a\begin{pmatrix}1\\2\\-1\\-2\end{pmatrix}+b\begin{pmatrix}2\\3\\0\\-1\end{pmatrix}+c\begin{pmatrix}1\\2\\1\\3\end{pmatrix}$$

for at least one of a,b,c different from zero. Since we must have

$$-a+c=-1,\quad -2a-b+3c=0$$

we can write $a=1+c$, $b=c-2$. We are now left with

$$\begin{aligned}a+2b+c=1,\quad c+1+2(c-2)+c=1,\quad 4c=4\\2a+3b+2c=3,\quad 2(c+1)+3(c-2)+2c=3,\quad 7c=7\end{aligned}$$

both of which have solution $c=1$. Then $a=2,\ b=-1$.

It is easy to show that the first three columns are independent. Hence the columns of A span a subspace of $\mathbb{R}^4$. Therefore it is possible that the vector given cannot be expressed as a linear combination of the columns of A. After some rather tedious calculations that's what we find.

Problem 2.7

Linear independence requires that

$$a\begin{pmatrix}1\\1\\1\end{pmatrix}+b\begin{pmatrix}1\\1\\2\end{pmatrix}+c\begin{pmatrix}1\\2\\3\end{pmatrix}=0\Rightarrow a=b=c=0.$$

From the first equation we have $c=-(a+b)$; then, for the other two equations,

$$a=2b,\qquad a=3b$$

which only has a solution if $a=b=0$. It follows that $c=0$. Next

$$x_1\begin{pmatrix}1\\1\\1\end{pmatrix}+x_2\begin{pmatrix}1\\1\\2\end{pmatrix}+x_3\begin{pmatrix}1\\2\\3\end{pmatrix}=\begin{pmatrix}6\\9\\14\end{pmatrix}$$

is solved for $x_1=1, x_2=2, x_3=3$.

Problem 2.8

Linear independence requires

$$a\begin{pmatrix}1\\2\\-1\\-2\end{pmatrix}+b\begin{pmatrix}2\\3\\0\\-1\end{pmatrix}+c\begin{pmatrix}1\\2\\1\\4\end{pmatrix}+d\begin{pmatrix}1\\3\\-1\\0\end{pmatrix}=0\Rightarrow a=b=c=d=0.$$

Use

$$2(a+2b+c+d)-(2a+3b+2c+3d)=0\Rightarrow b=d$$

to rewrite the condition as

$$a\begin{pmatrix}1\\2\\-1\\-2\end{pmatrix}+b\begin{pmatrix}3\\6\\-1\\-1\end{pmatrix}+c\begin{pmatrix}1\\2\\1\\4\end{pmatrix}=0$$

Next

$$(a+3b+c)+(-a-b+c)=0\Rightarrow b=-c$$

and

$$a\begin{pmatrix}1\\2\\-1\\-2\end{pmatrix}+b\begin{pmatrix}2\\4\\-2\\-5\end{pmatrix}=0$$

It is clear that $(1,2,-1,-2)$ and $(2,4,-2,-5)$ are independent. Hence $a=b=0$ and, from the last two equations $c=d=0$.

The coordinates of x are $(0,2,1,2)$

Problem 2.9

The hint given uses the fact that if two vectors are orthogonal then they are independent. Since the rows of H^T contain the columns of H, independence requires that $H^\mathsf{T}H$ is a diagonal matrix:

$$\begin{bmatrix}1&1&1&1\\1&1&-1&-1\\1&-1&0&0\\0&0&1&-1\end{bmatrix}\begin{bmatrix}1&1&1&0\\1&1&-1&0\\1&-1&0&1\\1&-1&0&-1\end{bmatrix}=\begin{bmatrix}4&0&0&0\\0&4&0&0\\0&0&2&0\\0&0&0&2\end{bmatrix}$$

Problem 2.10

As with Problem 2.9.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Problem 2.11

1. The condition for linear independence is

$$b_1 v_1 + \cdots + b_m v_m = 0 \Rightarrow b_1 = \cdots = b_n = 0.$$

Rewrite the linear combination as

$$\sum_{i=1}^{m} b_i v_i = \sum_{i=1}^{m} b_i \sum_{j=1}^{m} a_{ij} u_j = 0$$

Next, interchange the order of summation, and use the linear independence of u vectors to produce the following set of conditions:

$$\sum_{i=1}^{m} b_i a_{ij} = 0, \quad j = 1, \ldots, m. \tag{$*$}$$

Each column of A is a vector a_j; then given j the above sum is the inner product of a vector b with a_j. Since the inner product of vector b with all columns of A is zero, $b \notin \operatorname{span}(a_1, \ldots, a_m)$. If vectors $a_1, \ldots, a_m$ are independent then

$$c_1 a_1 + \cdots + c_m a_m = 0 \Rightarrow c_1 = \cdots = c_m = 0.$$

Note that since A is upper triangular we can write this as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ 0 & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{mm} \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

It is clear that $c_m = 0$; then we have

$$c_{m-1} a_{m-1,m-1} + c_m a_{m-1,m} = 0, \Rightarrow c_{m-1} = 0.$$

Continuing in this fashion we find that $c_1 = \cdots = c_m = 0$ and so the columns of A are independent vectors. There are m of them so they span the whole vector space. So it is not possible for vector b to be orthogonal to all a vectors. Hence we conclude that $(*)$ only holds if $b = 0$ and vectors $v_1, \cdots, v_m$ are independent.

2. We have already shown that, if the diagonal entries of A are nonzero, the columns of A are independent vectors. Therefore with the exception of the zero vector there is no vector x such that $Ax = 0$. For any vector y there is a unique vector x such that $y = Ax$. Otherwise, if $y = Ax'$ we must have $A(x' - x) = 0$ which leads to a contradiction. Therefore the inverse of A exists. Given $Ax = y$ then there is a simple recursive formula for the solutions of this system of equations:

$$\begin{aligned} x_m &= \frac{1}{a_{m,m}} y_m \\ x_{m-1} &= \frac{1}{a_{m-1,m-1}} \left(y_{m-1} - a_{m-1,m} x_m \right) \end{aligned}$$

$$\vdots$$

$$x_i = \frac{1}{a_{i,i}}\left(y_i - \sum_{k=i+1}^{m} a_{ik}x_k\right) \tag{$*$}$$

We also have $x = A^{-1}y$ and so

$$x_i = \sum_{k=1}^{m} a'_{ik}y_k.$$

In $(*)$ every x_k is only a function of y_l with $l \geq k$. Therefore x_i is only a function of $y_i, y_{i+1}, \ldots, y_m$ and $a'_{ik} = 0$ if $k < i$. Hence the inverse is also upper-triangular. From this we also conclude that if the diagonal elements of A are equal to zero then A^{-1} does not have an inverse.

3. We have already shown that if $(u_1, \ldots, u_m)$ are linearly independent then $(v_1, \ldots, v_m)$ are linearly independent. Now assume that $(v_1, \ldots, v_m)$ are linearly independent. If

$$v_i = \sum_{j=1}^{m} a_{ij}u_j$$

then in matrix form we have

$$\left[\begin{array}{c|c|c|c} v_1 & v_2 & \cdots & v_m \end{array}\right] = \left[\begin{array}{c|c|c|c} u_1 & u_2 & \cdots & u_m \end{array}\right] \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{mm} \end{bmatrix}$$

or, in shorthand,

$$V = UA^{\mathsf{T}}.$$

Since A has an inverse

$$U = V(A^{-1})^{\mathsf{T}}.$$

Therefore we can write

$$u_i = a'_{i1}v_1 + \cdots + a'_{im}v_m$$

where a'_{ij} are the elements of A^{-1}. From this point follow the steps in (1) to show that the u vectors are linearly independent given that the v vectors are independent.

Problem 2.12

1. We have a simple system of recursive equations with the following solution:

$$\begin{aligned} x_n &= b_n \\ x_{n-1} &= b_{n-1} - 2x_n = b_{n-1} - 2b_n \\ &\vdots \\ x_i &= b_i - 2x_{i+1} = b_i - 2b_{i+1} + 4b_{i+2} + \cdots + (-2)^{n-i}b_n \end{aligned}$$

2. From the solution above it is clear that if $b = 0$ then $x = 0$ and so since $\ker A = \{0\}$ it has an inverse. The inverse is an upper-triangular matrix:

$$A^{-1} = \begin{bmatrix} 1 & -2 & \cdots & (-2)^{n-2} & (-2)^{n-1} \\ 0 & 1 & \cdots & (-2)^{n-3} & (-2)^{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -2 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}.$$

If $n \geq 300$ then the largest coefficient is greater or equal to 2^{300}. As a comparison with the number of atoms in the universe, $10 \approx 2^3$ so $2^{300} \approx 10^{100}$.

3. Starting with

$$A(x + \Delta x) = b + \Delta b$$

since $Ax = b$ we have

$$\Delta x = A^{-1}\Delta b.$$

If $b = (0, 0, \ldots, 0, \epsilon)$ then, using the recursive formula,

$$(\Delta x)_1 = (-2)^{n-1}\epsilon$$

and

$$|(\Delta x)_1| = (2)^{n-1}|\epsilon|.$$

4.

shorthand: if e_i is a basis vector in $\mathbb{R}^n$ with all zero elements except from 1 at the ith element then the basis vectors of $M_{n,n}$ can be written as the outer product of two basis vectors (e.g. $E_{13} = e_1 e_3^\mathsf{T}$). From this we obtain the simple rule

$$E_{ij}E_{jk} = e_i e_j^\mathsf{T} e_j e_k^\mathsf{T} = e_i e_k^\mathsf{T} = E_{ik}$$

and $E_{ij}E_{kl} = O$ when $j \neq k$.

$$\begin{aligned}
A - I &= 2E_{12} + 2E_{23} + \cdots + 2E_{n-2,n-1} + 2E_{n-1,n} \\
(A - I)^2 &= 4E_{13} + 4E_{24} + \cdots + 4E_{n-3,n-1} + 4E_{n-2,n} \\
&\vdots \\
(A - I)^{n-1} &= 2^{n-1}E_{1,n} \\
(A - I)^n &= (2E_{12} + 2E_{23} + \cdots + 2E_{n-2,n-1} + 2E_{n-1,n})E_{1,n} = O
\end{aligned}$$

longhand:

$$A - I = \begin{bmatrix} 0 & 2 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

$$(A - I)^2 = \begin{bmatrix} 0 & 0 & 2^2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 2^2 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 2^2 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

$$\vdots$$

$$(A - I)^{n-1} = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 2^{n-1} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

$$(A-I)^n = O$$

Problem 2.13

1. First note that in the same way that

$$a^r - b^r = (a-b)(a^{r-1} + a^{r-2}b + \cdots + ab^{r-2} + b^{r-1})$$

where $a, b \in K$, it is also the case that

$$I^r - N^r = (I-N)(I^{r-1} + I^{r-1}N + \cdots + IN^{r-2} + N^{r-1}).$$

Since $N^r = O$ and all powers of I are equal to I we have

$$I = (I-N)(I + N + \cdots + N^{r-2} + N^{r-1})$$

Since

$$(I-N)(I-N)^{-1} = I$$

we conclude that

$$(I-N)^{-1} = I + N + \cdots + N^{r-2} + N^{r-1}.$$

2.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 3 & 4 & 5 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

or $A = I + N$.

$$N^2 = \begin{bmatrix} 0 & 0 & 4 & 12 & 25 \\ 0 & 0 & 0 & 4 & 12 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$N^3 = \begin{bmatrix} 0 & 0 & 0 & 8 & 36 \\ 0 & 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$N^4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 16 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$N^5 = O \Rightarrow r = 5$$

Therefore

$$\begin{aligned} A^{-1} = (I+N)^{-1} &= I - N + N^2 - N^3 + N^4 \\ &= \begin{bmatrix} 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Problem 2.14

1. If a matrix A is invertible then for every y there is a unique x such that $y = Ax$. Now assume that for a vector $x' \neq 0$, we have $Ax' = 0$. Then $y = Ax + 0 = Ax + Ax' = A(x + x')$. Since there is no unique x such that $y = Ax$ this matrix is not invertible. Only if $Ax = 0 \Rightarrow x = 0$ the matrix is invertible.
2. $I_m - AB$ is invertible if

$$(I_m - AB)y = 0 \Rightarrow y = 0.$$

We are given that

$$(I_n - BA)x = 0 \Rightarrow x = 0.$$

Multiply both sides with A (this is allowed since $A \in M_{m,n}$):

$$A(I_n - BA)x = 0, Ax - (AB)(Ax) = 0, (I_m - AB)(Ax) = 0.$$

Since $x = 0$, $Ax = 0$ and so $I_m - AB$ is invertible.

Problem 2.15

1. For

$$(n-3)b_1 - (b_2 + \cdots + b_n)$$

the first element of the resulting vector is

$$(n-3) \cdot 1 - (n-1) \cdot (-1) = 2n - 4 = 2(n-2).$$

For all other elements we have

$$(n-3) \cdot 1 - (n-2) \cdot 1 - (-1) = (n-3) - (n-3) = 0.$$

For $b_1 - b_i$ we always have the first element equal to

$$1 - (-1) - 2$$

while for all elements, except from the ith element,

$$1 - (1) = 0.$$

For the $i$the element we have

$$1 - (-1) = 2$$

and so $b_1 - b_i = 2(e_1 + e_i)$.
2. The equations in (1) can be written in matrix form:

$$B \begin{bmatrix} n-3 & 1 & 1 & \cdots & 1 & 1 & 1 \\ -1 & -1 & 0 & \cdots & 0 & 0 & 0 \\ -1 & 0 & -1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ -1 & 0 & 0 & \cdots & -1 & 0 & 0 \\ -1 & 0 & 0 & \cdots & 0 & -1 & 0 \\ -1 & 0 & 0 & \cdots & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 2(n-2) & 2 & 2 & \cdots & 2 & 2 & 2 \\ 0 & 2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 2 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 2 \end{bmatrix}$$

Remove the factor of 2 from the r.h.s.:

$$B\frac{1}{2} \begin{bmatrix} n-3 & 1 & 1 & \cdots & 1 & 1 & 1 \\ -1 & -1 & 0 & \cdots & 0 & 0 & 0 \\ -1 & 0 & -1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ -1 & 0 & 0 & \cdots & -1 & 0 & 0 \\ -1 & 0 & 0 & \cdots & 0 & -1 & 0 \\ -1 & 0 & 0 & \cdots & 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} (n-2) & 1 & 1 & \cdots & 1 & 1 & 1 \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

The matrix in the r.h.s. can be written in block form as

$$\begin{bmatrix} (n-2) & e^{\mathsf{T}} \\ 0 & I_{n-1} \end{bmatrix}$$

where $e^{\mathsf{T}} = (1, 1, \ldots, 1)$ (0 is a vector of $n-1$ zeros). It is easy to check that

$$\begin{bmatrix} (n-2) & e^{\mathsf{T}} \\ 0 & I_{n-1} \end{bmatrix} \begin{bmatrix} \frac{1}{(n-2)} & -\frac{1}{n-2}e^{\mathsf{T}} \\ 0 & I_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 0^{\mathsf{T}} \\ 0 & I_{n-1} \end{bmatrix}.$$

Therefore

$$\begin{bmatrix} \frac{1}{(n-2)} & -\frac{1}{(n-2)} & -\frac{1}{(n-2)} & \cdots & -\frac{1}{(n-2)} & -\frac{1}{(n-2)} & -\frac{1}{(n-2)} \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

is the inverse of the matrix in the r.h.s. The inverse of B is equal to the following product:

$$\frac{1}{2} \begin{bmatrix} n-3 & 1 & 1 & \cdots & 1 & 1 & 1 \\ -1 & -1 & 0 & \cdots & 0 & 0 & 0 \\ -1 & 0 & -1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ -1 & 0 & 0 & \cdots & -1 & 0 & 0 \\ -1 & 0 & 0 & \cdots & 0 & -1 & 0 \\ -1 & 0 & 0 & \cdots & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{(n-2)} & -\frac{1}{(n-2)} & -\frac{1}{(n-2)} & \cdots & -\frac{1}{(n-2)} & -\frac{1}{(n-2)} & -\frac{1}{(n-2)} \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

Simplify by writing both matrices in block form

$$\begin{bmatrix} \frac{n-3}{2} & \frac{1}{2}e^{\mathsf{T}} \\ -\frac{1}{2}e & -\frac{1}{2}I_{n-1} \end{bmatrix} \begin{bmatrix} \frac{1}{n-2} & -\frac{1}{n-2}e^{\mathsf{T}} \\ 0 & I_{n-1} \end{bmatrix} = \begin{bmatrix} \frac{n-3}{2(n-2)} & -\frac{n-3}{2(n-2)}e^{\mathsf{T}} + \frac{1}{2}e^{\mathsf{T}} \\ -\frac{1}{2(n-2)}e & \frac{1}{2(n-2)}ee^{\mathsf{T}} - \frac{1}{2}I_{n-1} \end{bmatrix}$$

Since

$$-\frac{n-3}{2(n-2)} + \frac{1}{2} = \frac{-(n-3)+(n-2)}{2(n-2)} = \frac{1}{2(n-2)}$$

the off-diagonal elements in the first row are equal to this value. Next, since ee^{T} is a matrix of ones,

$$\frac{1}{2(n-2)}ee^{\mathsf{T}} - \frac{1}{2}I_{n-1}$$

is a matrix with off-diagonal elements equal to $1/2(n-2)$ and diagonal elements equal to

$$\frac{1}{2(n-2)} - \frac{1}{2} = \frac{1-(n-2)}{2(n-2)} = -\frac{n-3}{2(n-2)}.$$

3. Define $D_i = \text{diag}(b_i)$ where diag converts a vector into a diagonal matrix. First, assume that the columns of B are independent. Then since any $n \times n$ diagonal matrix is $\text{diag}(x)$ where $x \in \mathbb{R}^n$ we can write

$$\text{diag}(x) = \text{diag}(x_1 b_1 + \cdots x_n b_n).$$

Since $x_1, \ldots, x_n$ are scalars and diagonal matrices form a vector space we have

$$\text{diag}(x) = x_1 \,\text{diag}(b_1) \cdots + x_n \,\text{diag}(b_n).$$

Hence the diagonal matrices generated using the columns of B can be used as a basis for the vector space of diagonal matrices. Independence of the columns of B follows from the fact that B has an inverse.

Problem 2.16

$$(A^k B_k)_{ij} = a_{ik} b_{kj}$$

where a_{ik} (resp. b_{kj}) are the elements of of A (resp. B). Hence

$$\sum_{k=1}^{n} (A^k B_k)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

From the definition of the matrix product

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

and so

$$\sum_{k=1}^{n} (A^k B_k)_{ij} = (AB)_{ij}.$$

Problem 2.17

From the definition of the inverse

$$f^{-1} f(ax + by) = ax + by$$

where a, b are scalars and x, y are vectors. Since f is a linear map

$$f^{-1}(af(x) + bf(y)) = ax + by.$$

If f^{-1} is not linear then

$$f^{-1}(af(x) + bf(y)) \neq af^{-1} f(x) + bf^{-1} f(y) = ax + by.$$

We have arrived at a contradiction and this completes the proof.

Problem 2.18

For any $x \in E$,

$$x = x_1 u_1 + \cdots + x_n u_n$$

where n is the dimension of E. From the definition of the linear map

$$f(x) = x_1 f(u_1) + \cdots + x_n f(u_n).$$

If $(f(u_i))_{i \in I}$ spans F then if we vary the coordinates $x_1, \ldots, x_n$ we can generate any vector in F. Hence f is surjective.

Problem 2.19

We can generate an orthonormal basis of E using vectors $v/\|v\|, e_1, \ldots, e_{n-1}$ ($\|v\|$ is the norm of vector v; vectors $e_1, \ldots, e_{n-1}$ are not the canonical basis vectors). Hence $\dim \ker f = n-1$ since

$$Ae_j = uv^\mathsf{T} e_j = 0, \quad \text{for } j = 1, \ldots, n$$

and

$$Av = uv^\mathsf{T} \frac{v}{\|v\|} = \|v\| u \neq 0.$$

So f has rank 1.

Problem 2.20

$$\phi_1 - 10\phi_2 + 7\phi_3 = 0, \quad \text{for all } x, y, z.$$

since

$$\begin{bmatrix} 2 & 3 & 4 \\ -1 & -5 & -7 \\ 3 & 1 & 1 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

if

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ -10 \\ 7 \end{pmatrix}.$$

Problem 2.21

ϕ_1, ϕ_2, ϕ_3 are linearly independent if

$$a\phi_1 + b\phi_2 + c\phi_3 = 0 \Rightarrow a = b = c = 0.$$

To show this write the following system of equations:

$$\begin{aligned} a + 2b + 3c &= 0 \\ 2a + 3b + 7c &= 0 \\ a + 3b + c &= 0 \end{aligned}$$

With $c = -a - 3b$ the first two equations are

$$\begin{aligned} -2a - 7b &= 0 \\ -5a - 18b &= 0 \end{aligned}$$

The only solution is $a = b = 0$; then $c = 0$ and the three linear forms are independent. It is easy to show that

$$\phi = -6\phi_1 + 2\phi_2 + \phi_3.$$

Chapter 3

Problem 3.22

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 7 \\ 1 & 3 & 1 \end{bmatrix} \xrightarrow{-2R1+R2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 3 & 1 \end{bmatrix} \xrightarrow{-R1+R3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\xrightarrow{-R2}\begin{bmatrix}1&2&3\\0&1&-1\\0&1&-2\end{bmatrix}\xrightarrow{-R2+R3}\begin{bmatrix}1&2&3\\0&1&-1\\0&0&-1\end{bmatrix}\xrightarrow{-R3}\begin{bmatrix}1&2&3\\0&1&-1\\0&0&1\end{bmatrix}.$$

Some high school math:

$$-27\begin{bmatrix}1\\2\\1\end{bmatrix}+9\begin{bmatrix}2\\3\\3\end{bmatrix}+4\begin{bmatrix}3\\7\\1\end{bmatrix}=\begin{bmatrix}3\\1\\4\end{bmatrix}$$
$$-61\begin{bmatrix}1\\2\\1\end{bmatrix}+18\begin{bmatrix}2\\3\\3\end{bmatrix}+10\begin{bmatrix}3\\7\\1\end{bmatrix}=\begin{bmatrix}5\\2\\3\end{bmatrix}$$
$$-41\begin{bmatrix}1\\2\\1\end{bmatrix}+9\begin{bmatrix}2\\3\\3\end{bmatrix}+8\begin{bmatrix}3\\7\\1\end{bmatrix}=\begin{bmatrix}1\\1\\-6\end{bmatrix}$$

Apply the following operation on P_1:

$$\begin{bmatrix}-27&-61&-41\\9&18&9\\4&10&8\end{bmatrix}\xrightarrow{-2C1+C2\to C3}\begin{bmatrix}-27&-61&-7\\9&18&0\\4&10&2\end{bmatrix}$$

Now

$$A_1P_1=\begin{bmatrix}3&5&-1\\1&2&0\\4&3&-5\end{bmatrix}\neq B_1.$$

It is easy to check that B_1 is not invertible since $C3=C2-2C1$.

Problem 3.23

Proof that A_2 is invertible:

$$\begin{bmatrix}1&1&1&1\\1&2&1&3\\1&1&2&2\\1&1&1&3\end{bmatrix}\xrightarrow{-R1+R2}\begin{bmatrix}1&1&1&1\\0&1&0&2\\1&1&2&2\\1&1&1&3\end{bmatrix}\xrightarrow{-R1+R3}\begin{bmatrix}1&1&1&1\\0&1&0&2\\0&0&1&1\\1&1&1&3\end{bmatrix}$$
$$\xrightarrow{-R1+R4}\begin{bmatrix}1&1&1&1\\0&1&0&2\\0&0&1&1\\0&0&0&2\end{bmatrix}\xrightarrow{\frac{1}{2}R4}\begin{bmatrix}1&1&1&1\\0&1&0&2\\0&0&1&1\\0&0&0&1\end{bmatrix}.$$

Proof that B_2 is invertible:

$$\begin{bmatrix}1&-2&2&-2\\0&-3&2&-3\\3&-5&5&-4\\3&-4&4&-4\end{bmatrix}\xrightarrow{-3R1+R3}\begin{bmatrix}1&-2&2&-2\\0&-3&2&-3\\0&1&-1&2\\3&-4&4&-4\end{bmatrix}\xrightarrow{-3R1+R4}\begin{bmatrix}1&-2&2&-2\\0&-3&2&-3\\0&1&-1&2\\0&2&-2&2\end{bmatrix}$$
$$\xrightarrow{-\frac{1}{3}R2}\begin{bmatrix}1&-2&2&-2\\0&1&-\frac{2}{3}&1\\0&1&-1&2\\0&2&-2&2\end{bmatrix}\xrightarrow{-R2+R3}\begin{bmatrix}1&-2&2&-2\\0&1&-\frac{2}{3}&1\\0&0&-\frac{1}{3}&1\\0&2&-2&2\end{bmatrix}\xrightarrow{-2R2+R4}\begin{bmatrix}1&-2&2&-2\\0&1&-\frac{2}{3}&1\\0&0&-\frac{1}{3}&1\\0&0&-\frac{2}{3}&0\end{bmatrix}$$
$$\xrightarrow{-3R3}\begin{bmatrix}1&-2&2&-2\\0&1&-\frac{2}{3}&1\\0&0&1&-4\\0&0&-\frac{2}{3}&0\end{bmatrix}\xrightarrow{\frac{2}{3}R3+R4}\begin{bmatrix}1&-2&2&-2\\0&1&-\frac{2}{3}&1\\0&0&1&-4\\0&0&0&1\end{bmatrix}.$$

To prove that the coordinates of the column vectors of matrix B_2 over the basis consisting of the column vectors of A_2 are the columns of P_2 do a check like

$$2\begin{pmatrix}1\\1\\1\\1\end{pmatrix} - 3\begin{pmatrix}1\\2\\1\\1\end{pmatrix} + \begin{pmatrix}1\\1\\2\\1\end{pmatrix} + \begin{pmatrix}1\\3\\2\\3\end{pmatrix} = \begin{pmatrix}1\\0\\3\\3\end{pmatrix}$$

Repeat the same calculations for all other columns of B_2. One takeaway from this calculations is that the individual operations using a column of P_2 as the coordinates of a vector over the basis defined by the columns of A_2 is equivalent to the matrix multiplication A_2P_2. We have already shown that this is equal to B_2.

To prove that the value given for P_2^{-1} is correct check that

$$B_2P_2^{-1} = \begin{bmatrix}1 & -2 & 2 & -2\\0 & -3 & 2 & -3\\3 & -5 & 5 & -4\\3 & -4 & 4 & -4\end{bmatrix}\begin{bmatrix}-1 & -1 & -1 & 1\\2 & 1 & 1 & -2\\2 & 1 & 2 & -3\\-1 & -1 & 0 & -1\end{bmatrix} = \begin{bmatrix}1 & 1 & 1 & 1\\1 & 2 & 1 & 3\\1 & 1 & 2 & 2\\1 & 1 & 1 & 3\end{bmatrix}.$$

The mouthfull

> the coordinates over the basis consisting of the column vectors of B_2 of the vector whose coordinates over the basis consisting of the column vectors of A_2 are $(2, -3, 0, 0)$

can be summarized as

$$B_2x = A_2y$$

where

$$y^\mathsf{T} = \begin{pmatrix}2 & -3 & 0 & 0\end{pmatrix}.$$

Since $B_2 = A_2P_2$ write this as

$$P_2x = y, \quad x = P_1^{-1}y.$$

We have

$$x = \begin{bmatrix}-1 & -1 & -1 & 1\\2 & 1 & 1 & -2\\2 & 1 & 2 & -3\\-1 & -1 & 0 & -1\end{bmatrix}\begin{pmatrix}2\\-3\\0\\0\end{pmatrix} = \begin{pmatrix}1\\1\\1\\1\end{pmatrix}.$$

Problem 3.24

1. To show that

$$\begin{pmatrix}B_0^2(t)\\B_1^2(t)\\B_2^2(t)\end{pmatrix} = \begin{bmatrix}1 & -2 & 1\\0 & 2 & -2\\0 & 0 & 1\end{bmatrix}\begin{pmatrix}1\\t\\t^2\end{pmatrix}$$

is trivial as it only requires polynomial expansion which any high school student should be able to do. Since

$$B_0^2(t) + B_1^2(t) + B_2^2(t) = \begin{pmatrix}1 & 1 & 1\end{pmatrix}\begin{pmatrix}B_0^2(t)\\B_1^2(t)\\B_2^2(t)\end{pmatrix}$$

it suffices to show that

$$\begin{pmatrix}1 & 1 & 1\end{pmatrix}\begin{bmatrix}1 & -2 & 1\\0 & 2 & -2\\0 & 0 & 1\end{bmatrix} = \begin{pmatrix}1 & 0 & 0\end{pmatrix}.$$

2. For the first row of the matrix use the formula for the binomial expansion:

$$(1-t)^3 = 1 - 3t + 3t^2 - t^3.$$

More high school algebra produces the other rows.

Following the same procedure as in (1) we can show that

$$\begin{pmatrix}1 & 1 & 1 & 1\end{pmatrix}\begin{bmatrix}1 & -3 & 3 & -1\\ 0 & 3 & -6 & 3\\ 0 & 0 & 3 & -3\\ 0 & 0 & 0 & 1\end{bmatrix} = \begin{pmatrix}1 & 0 & 0 & 0\end{pmatrix}.$$

3. For this proof start with the assertion that $1, t, t^2$ are linearly independent. This means that

$$a_0 + a_1 t + a_2 t^2 = 0 \quad \text{for all } t \quad \Rightarrow a_0 = a_1 = a_2 = 0.$$

Since $a_0 + a_1 t + a_2 t^2$ is a quadratic equation it can have only two solutions (ignore whether $K = \mathbb{C}$ or $\mathbb{R}$). The condition for linear independence requires that the quadratic must have an infinite number of solutions. This is only possible if $a_0 = a_1 = a_2 = 0$ completing the proof.

Since the Bernstein polynomials are linear combinations of the basis, linear independence requires that

$$\begin{bmatrix}1 & -2 & 1\\ 0 & 2 & -2\\ 0 & 0 & 1\end{bmatrix}$$

has an inverse. This is clear since the matrix is upper-triangular with non-zero diagonal elements. The same procedure provides proof for Bernstein polynomials of degree 3.

Problem 3.25

1. Use

$$(1-t)^{m-k} = \sum_{r=0}^{m-k} \binom{m-k}{r} (-t)^{m-k-r}$$

and

$$\binom{m-k}{r}\binom{m}{k} = \binom{m}{k+r}\binom{k+r}{k}$$

together with the change of variable $j = k + r$ to produce (3.5).

Following the same procedure as above it is easy to show that $1, t, t^2, \ldots, t^m$ are linearly independent. From (3.5)

$$\begin{pmatrix} B_0^m(t)\\ B_1^m(t)\\ \vdots\\ B_{m-1}^m(t)\\ B_m^m(t)\end{pmatrix} = \begin{bmatrix}1 & -m & \cdots & (-1)^{m-1}m & (-1)^m m\\ 0 & m & \cdots & (-1)^{m-2}m(m-1) & (-1)^{m-1}m\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & \cdots & m & -m\\ 0 & 0 & \cdots & 0 & 1\end{bmatrix}\begin{pmatrix}1\\ t\\ \vdots\\ t^{m-1}\\ t^m\end{pmatrix}$$

The matrix in the r.h.s. is upper-triangular with non-zero diagonal elements. Hence it is invertible and so the elements of the vector in the l.h.s. also constitute a basis for the vector space of polynomials with degree up to m. Therefore $B_0^m(t), B_1^m(t), \ldots, B_{m-1}^m(t), B_m^m(t)$ are linearly independent.

2.

$$B_0^m(t) + B_1^m(t) + \cdots + B_{m-1}^m(t) + B_m^m(t) = \begin{pmatrix}1 & 1 & \cdots & 1 & 1\end{pmatrix}\begin{pmatrix} B_0^m(t)\\ B_1^m(t)\\ \vdots\\ B_{m-1}^m(t)\\ B_m^m(t)\end{pmatrix}$$

Next note that to resolve

$$\begin{pmatrix}1 & 1 & \cdots & 1 & 1\end{pmatrix}\begin{bmatrix}1 & -m & \cdots & (-1)^{m-1}m & (-1)^m m\\ 0 & m & \cdots & (-1)^{m-2}m(m-1) & (-1)^{m-1}m\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & \cdots & m & -m\\ 0 & 0 & \cdots & 0 & 1\end{bmatrix}$$

we must evaluate sums like

$$S_j = \sum_{k=0}^{m}(-1)^{j-k}\binom{m}{j}\binom{j}{k}$$

each keeping j fixed. For $j = 0$ it is easy to show that $S_0 = 1$; for all other values of j

$$\sum_{k=0}^{j}(-1)^{j-k}\binom{m}{j}\binom{j}{k} = \binom{m}{j}\sum_{k=0}^{j}(-1)^{j-k}\binom{j}{k} = \binom{m}{j}\sum_{k=0}^{j}(-1)^{j-k}1^k\binom{j}{k} = \binom{m}{j}(1-1)^j = 0.$$

Hence

$$\begin{pmatrix}1 & 1 & \cdots & 1 & 1\end{pmatrix}\begin{bmatrix}1 & -m & \cdots & (-1)^{m-1}m & (-1)^m m\\ 0 & m & \cdots & (-1)^{m-2}m(m-1) & (-1)^{m-1}m\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & \cdots & m & -m\\ 0 & 0 & \cdots & 0 & 1\end{bmatrix} = \begin{pmatrix}1 & 0 & \cdots & 0 & 0\end{pmatrix}$$

and the proof is complete.

3. We have already shown that this matrix is upper-triangular. The diagonal elements are equal to $\binom{m}{k}$. This matrix is symmetric w.r.t. the anti-diagonal. This is easy to see for the diagonal elements; for all other elements note that element (k, j) is reflected across the anti-diagonal to element $(m-j, m-k)$. Then the value of the reflected element is

$$\begin{aligned}(-1)^{m-k-(m-j)}\binom{m}{m-k}\binom{m-k}{m-j} &= (-1)^{j-k}\frac{m!}{(m-k)!k!}\frac{(m-k)!}{(m-j)!(j-k)!}\\ &= (-1)^{j-k}\frac{m!}{j!k!}\frac{j!}{(m-j)!(j-k)!} = (-1)^{j-k}\frac{m!}{j!(m-j)!}\frac{j!}{k!(j-k)!}\\ &= (-1)^{j-k}\binom{m}{j}\binom{j}{k}\end{aligned}$$

which is equal to the value of the original element.

We have already shown that, with the exception of column 1, the sum of the elements for all other columns is equal to zero. For the sum of the elements along each row note that for $t = 1$ all elements of the basis are equal to 1. Therefore the sum of the elements for row k is equal to $B_k^m(1) = 0$; the exception to this rule is row m (using zero indexing) for which the sum equals 1.

4. Using the definition of a Bernstein polynomial

$$\sum_{j=0}^{m-i} t^i B_j^{m-i}(t) = \sum_{j=0}^{m-i} t^i\binom{m-i}{j}(1-t)^{m-i-j}t^j = t^i\sum_{j=0}^{m-i}\binom{m-i}{j}(1-t)^{m-i-j}t^j = t^i(1-t+t)^{m-i} = t^i.$$

Next

$$\begin{aligned}\binom{m}{i}\binom{m-i}{j} &= \frac{m!}{i!(m-i)!}\frac{(m-i)!}{j!(m-i-j)!} = \frac{m!}{i!(i+j)!}\frac{(i+j)!}{j!(m-i-j)!}\\ &= \frac{m!}{(m-i-j)!(i+j)!}\frac{(i+j)!}{i!j!} = \binom{m}{i+j}\binom{i+j}{i}.\end{aligned}$$

Using these facts

$$\begin{aligned}\sum_{j=0}^{m-i}\frac{\binom{i+j}{i}}{\binom{m}{i}}B_{i+j}^m(t) &= \sum_{j=0}^{m-i}\frac{\binom{i+j}{i}}{\binom{m}{i}}\binom{m}{i+j}(1-t)^{m-i-j}t^{i+j}\\ &= \sum_{j=0}^{m-i}\binom{m-i}{j}(1-t)^{m-i-j}t^{i+j}\\ &= \sum_{j=0}^{m-i}t^i\binom{m-i}{j}(1-t)^{m-i-j}t^j\\ &= \sum_{j=0}^{m-i}t^iB_j^{m-i}(t) = t^i.\end{aligned}$$

We can write

$$\begin{pmatrix}1\\ t\\ \vdots\\ t^{m-1}\\ t^m\end{pmatrix} = \begin{bmatrix}1 & 1 & \cdots & 1 & 1\\ 0 & \frac{1}{m} & \cdots & \frac{m-1}{m} & 1\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & \cdots & \frac{1}{m} & 1\\ 0 & 0 & \cdots & 0 & 1\end{bmatrix}\begin{pmatrix}B_0^m(t)\\ B_1^m(t)\\ \vdots\\ B_{m-1}^m(t)\\ B_m^m(t)\end{pmatrix}$$

The matrix is upper-triangular with non-zero diagonal elements and so it is invertible. Hence the Bernstein polynomials form a basis of the vector space of polynomials of degree $\leq m$.

From the explicit formulas given in the matrix above it easy to derive its elements for $m = 2, 3$.

5. The easiest way to write this is to use the shorthand

$$x = \alpha^\mathsf{T} p_1, y = \beta^\mathsf{T} p_2$$

where

$$\alpha^\mathsf{T} = (\alpha_{m_1}, \ldots, \alpha_1, \alpha_0), p_1^\mathsf{T} = (1, t, \ldots, t^{m_1-1}, t^{m_1}), \beta^\mathsf{T} = (\beta_{m_2}, \ldots, \beta_1, \beta_0), p_2^\mathsf{T} = (1, t, \ldots, t^{m_2-1}, t^{m_2}).$$

We know that

$$p_1 = B_1 b$$

where

$$b^\mathsf{T} = (B_0^m, \ldots, B_{m-1}^m, B_m^m)$$

and B_1 is the matrix whose explicit form we derived in (4) (since $m_1 \leq m$ it would be a $(m_1 + 1) \times (m + 1)$ submatrix of B). In the same way

$$p_2 = B_2 b.$$

where B_2 is an $(m_2 + 1) \times (m + 1)$ submatrix of B. Hence

$$x = \alpha^\mathsf{T} B_1 b, \qquad y = \beta^\mathsf{T} B_2 b.$$

The coefficients of x (resp. y) in terms of the Bernstein polynomials are given by $\alpha^\mathsf{T} B_1$ (resp. $\beta^\mathsf{T} B_2$). If we denote $(x(t), y(t))$ by $C(t)$ then any point on the curve is

$$C(t) = B_0^m(t)b_0 + B_1^m(t)b_1 + \cdots + B_m^m(t)b_m.$$

$b_0, b_1, \ldots, b_m$ are fixed points. We have already shown that

$$B_0^m(t) + B_1^m(t) + \cdots + B_m^m(t) = 1, \quad \text{for all } t.$$

and we know that $B_i^m(t) \geq 0$. Since $C(t)$ is a weighted average $b_0, b_1, \ldots, b_m$ these points are not generally on the curve. The only exception is b_0 and b_m since for $t = 0$, $B_0^m(0) = 1$, $B_k^m(0) = 0$ for $k > 1$, while for $t = 1$, $B_m^m(1) = 1$, $B_k^m(0) = 0$ for $k < m$.

Problem 3.26

Write A in block form

$$A = \begin{bmatrix} \mathbf{0}^\mathsf{T} & -a_n \\ \boldsymbol{I}_{n-1} & -\boldsymbol{a} \end{bmatrix}$$

where $\boldsymbol{a}^\mathsf{T} = (a_{n-1}, a_{n-2}, \ldots, a_2, a_1)$ (we use bold lower case for vector, bold upper case for matrices). We can write P two different ways

$$P = \begin{bmatrix} \boldsymbol{c}^\mathsf{T} & p_{1n} \\ \boldsymbol{Q} & \boldsymbol{q} \end{bmatrix}$$

or

$$P = \begin{bmatrix} \boldsymbol{b} & \boldsymbol{R} \\ p_{n1} & \boldsymbol{r}^\mathsf{T} \end{bmatrix}.$$

Note that $\boldsymbol{q}, \boldsymbol{r}$ are $(n-1) \times 1$ vectors with a common $n-1$th element. Also $\boldsymbol{Q}, \boldsymbol{R}$ share the central submatrix of P obtained by excluding rows $1, n$ and colums $1, n$. We must have

$$PA^\mathsf{T} = AP$$

or

$$\begin{bmatrix} \boldsymbol{c}^\mathsf{T} & p_{1n} \\ \boldsymbol{Q} & \boldsymbol{q} \end{bmatrix} \begin{bmatrix} \mathbf{0} & \boldsymbol{I} \\ -a_n & -\boldsymbol{a}^\mathsf{T} \end{bmatrix} = \begin{bmatrix} \mathbf{0}^\mathsf{T} & -a_n \\ \boldsymbol{I} & -\boldsymbol{a} \end{bmatrix} \begin{bmatrix} \boldsymbol{b} & \boldsymbol{R} \\ p_{n1} & \boldsymbol{r}^\mathsf{T} \end{bmatrix}.$$

This can be simplified to

$$\begin{bmatrix} -p_{1n}a_n & \boldsymbol{c}^\mathsf{T} - p_{1n}\boldsymbol{a}^\mathsf{T} \\ -a_n\boldsymbol{q} & \boldsymbol{Q} - \boldsymbol{q}\boldsymbol{a}^\mathsf{T} \end{bmatrix} = \begin{bmatrix} -p_{n1}a_n & -a_n\boldsymbol{r}^\mathsf{T} \\ \boldsymbol{b} - p_{n1}\boldsymbol{a} & \boldsymbol{R} - \boldsymbol{a}\boldsymbol{r}^\mathsf{T} \end{bmatrix}$$

We obtain the following conditions:

$$\begin{aligned} -p_{1n}a_n &= -p_{n1}a_n, \quad p_{1n} = p_{n1} \\ \boldsymbol{c}^\mathsf{T} - p_{1n}\boldsymbol{a}^\mathsf{T} &= -a_n\boldsymbol{r}^\mathsf{T} \\ \boldsymbol{b} - p_{n1}\boldsymbol{a} &= -a_n\boldsymbol{q} \\ \boldsymbol{Q} - \boldsymbol{q}\boldsymbol{a}^\mathsf{T} &= \boldsymbol{R} - \boldsymbol{a}\boldsymbol{r}^\mathsf{T} \end{aligned}$$

If we write the explicit form for $\boldsymbol{q}\boldsymbol{a}^\mathsf{T} - \boldsymbol{a}\boldsymbol{r}^\mathsf{T}$ we obtain

$$\begin{bmatrix} (p_{2n} - p_{n2})a_{n-1} & p_{2n}a_{n-2} - a_{n-1}p_{n3} & \cdots & p_{2n}a_2 - a_{n-1}p_{n,n-1} & p_{2n}a_1 - a_{n-1}p_{n,n} \\ p_3 n a_{n-1} - a_{n-2}p_{n2} & (p_{3n} - p_{n3})a_{n-2} & \cdots & p_{2n}a_2 - a_{n-2}p_{n,n-1} & p_{3n}a_1 - a_{n-2}p_{n,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ p_{n-1,n}a_{n-1} - a_2p_{n,2} & p_{n-1,n}a_{n-2} - a_2p_{n,3} & \cdots & (p_{n-1,n} - p_{n,n-1})a_2 & p_{n-1,n}a_1 - a_2p_{n,n} \\ p_{n,n}a_{n-1} - a_1p_{n,2} & p_{n,n}a_{n-2} - a_1p_{n,3} & \cdots & p_{n,n}a_2 - a_1p_{n,n-1} & 0 \end{bmatrix}.$$

From the construction of $\boldsymbol{Q}, \boldsymbol{R}$ we obtain differences of elements of P whose indices sum to the same total:

$$\begin{bmatrix} p_{21} - p_{12} & p_{22} - p_{13} & \cdots & p_{2,n-2} - p_{1,n-1} & p_{2,n-1} - p_{1,n} \\ p_{31} - p_{22} & p_{32} - p_{23} & \cdots & p_{3,n-2} - p_{2,n-1} & p_{3,n-1} - p_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ p_{n-1,1} - p_{n-2,2} & p_{n-1,2} - p_{n-2,3} & \cdots & p_{n-1,n-2} - p_{n-2,n-1} & p_{n-1,n-1} - p_{n-2,n} \\ p_{n,1} - p_{n-1,2} & p_{n,2} - p_{n-1,3} & \cdots & p_{n,n-2} - p_{n-1,n-1} & p_{n,n-1} - p_{n-1,n} \end{bmatrix}.$$

From the equality of $\boldsymbol{Q} - \boldsymbol{R}$ and $\boldsymbol{q}\boldsymbol{a}^\mathsf{T} - \boldsymbol{a}\boldsymbol{r}^\mathsf{T}$ we must have $p_{n,n-1} = p_{n-1,n}$. But then the penultimate diagonal element of $\boldsymbol{q}\boldsymbol{a}^\mathsf{T} - \boldsymbol{a}\boldsymbol{r}^\mathsf{T}$ must be zero and so we must have $p_{n-1,n-2} = p_{n-2,n-1}$. Following

the same procedure we obtain additional symmetric conditions for some elements of P. To show that all of P is symmetric we sum elements of $\boldsymbol{Q}-\boldsymbol{R}$. For example.

$$p_{n,n-2}-p_{n-1,n-1}+p_{n-1,n-1}-p_{n-2,n}$$

must be equal to

$$p_{n,n}a_2-a_1p_{n,n-1}+p_{n-1,n}a_1-a_2p_{n,n}=0.$$

Therefore we must have $p_{n,n-2}=p_{n-2,n}$. Following the same steps we conclude that P is symmetric. However, even with this restriction,

$$PA^{\mathsf{T}}=P^{\mathsf{T}}A^{\mathsf{T}}=AP$$

does not have a unique solution. Since A is invertible we can choose any symmetric matrix S and then $P=A^{-1}S$ is a solution. Also note that since

$$PA^{\mathsf{T}}=AP\Rightarrow(bP)A^{\mathsf{T}}=A(bP),\quad b\in K$$

this problem is **scale invariant**.

The conditions given are that all elements above the antidiagonal are zero while for p_{ij} where $i+j=n+k$ $(k\geq 0)$ all are equal. From this condition $\boldsymbol{Q}-\boldsymbol{R}=\boldsymbol{O}$ where $\boldsymbol{O}$ is the zero matrix. The same is true for all elements of $\boldsymbol{q}\boldsymbol{a}^{\mathsf{T}}-\boldsymbol{a}\boldsymbol{r}^{\mathsf{T}}$, so from the last row we obtain

$$p_{n,i}=\frac{p_{n,n}a_{n-i+1}}{a_1},\quad i=2,3,\ldots,n-1.$$

Repeating the same procedure for all other rows we obtain a matrix with the following structure:

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & \frac{p_{n,n}a_n}{a_1} \\ 0 & 0 & \cdots & \frac{p_{n,n}a_n}{a_1} & \frac{p_{n,n}a_{n-1}}{a_1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \frac{p_{n,n}a_n}{a_1} & \cdots & \frac{p_{n,n}a_3}{a_1} & \frac{p_{n,n}a_2}{a_1} \\ \frac{p_{n,n}a_n}{a_1} & \frac{p_{n,n}a_{n-1}}{a_1} & \cdots & \frac{p_{n,n}a_2}{a_1} & p_{n,n} \end{bmatrix}.$$

If we factor out $p_{n,n}/a_1$ we obtain the simpler form

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & a_n \\ 0 & 0 & \cdots & a_n & a_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_n & \cdots & a_3 & a_2 \\ a_n & a_{n-1} & \cdots & a_2 & a_1 \end{bmatrix}.$$

It is easy to check that this satisfies the similarity condition. For example if $n=5$,

$$\begin{aligned} PA^{\mathsf{T}} &= \begin{bmatrix} 0 & 0 & 0 & 0 & a_5 \\ 0 & 0 & 0 & a_5 & a_4 \\ 0 & 0 & a_5 & a_4 & a_3 \\ 0 & a_5 & a_4 & a_3 & a_2 \\ a_5 & a_4 & a_3 & a_2 & a_1 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -a_5 & -a_4 & -a_3 & -a_2 & -a_1 \end{bmatrix} \\ &= \begin{bmatrix} -a_5^2 & -a_4a_5 & -a_3a_5 & -a_2a_5 & -a_1a_5 \\ -a_4a_5 & -a_4^2 & -a_3a_4 & -a_2a_4 & -a_1a_4+a_5 \\ -a_3a_5 & -a_3a_4 & -a_3^2 & -a_2a_3+a_5 & -a_1a_3+a_4 \\ -a_2a_5 & -a_2a_4 & -a_2a_3+a_5 & -a_2^2+a_4 & -a_1a_2+a_3 \\ -a_1a_5 & -a_1a_4+a_5 & -a_1a_3+a_4 & -a_1a_2+a_3 & -a_1^2+a_2 \end{bmatrix} \end{aligned}$$

and

$$AP = \begin{bmatrix} 0 & 0 & 0 & 0 & -a_5 \\ 1 & 0 & 0 & 0 & -a_4 \\ 0 & 1 & 0 & 0 & -a_3 \\ 0 & 0 & 1 & 0 & -a_2 \\ 0 & 0 & 0 & 1 & -a_1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & a_5 \\ 0 & 0 & 0 & a_5 & a_4 \\ 0 & 0 & a_5 & a_4 & a_3 \\ 0 & a_5 & a_4 & a_3 & a_2 \\ a_5 & a_4 & a_3 & a_2 & a_1 \end{bmatrix}$$

$$= \begin{bmatrix} -a_5^2 & -a_4a_5 & -a_3a_5 & -a_2a_5 & -a_1a_5 \\ -a_4a_5 & -a_4^2 & -a_3a_4 & -a_2a_4 & -a_1a_4 + a_5 \\ -a_3a_5 & -a_3a_4 & -a_3^2 & -a_2a_3 + a_5 & -a_1a_3 + a_4 \\ -a_2a_5 & -a_2a_4 & -a_2a_3 + a_5 & -a_2^2 + a_4 & -a_1a_2 + a_3 \\ -a_1a_5 & -a_1a_4 + a_5 & -a_1a_3 + a_4 & -a_1a_2 + a_3 & -a_1^2 + a_2 \end{bmatrix}.$$

2. Since $P^{-1}P = I$ we want the rows of P^{-1} to form a dual basis to the columns of P. Since P has zero elements above the antidiagonal set the elements of P^{-1} below the antidiagonal to zero. The product of the last row of P^{-1} times the last column of P is $p'_{n1}a_n = 1$ and so $p'_{n1} = 1/a_n$ (the product of the last row o P^{-1} with all other columns of P is zero by construction). Then for the penultimate row of P^{-1} we have the following two conditions:

$$p'_{n-1,2}a_n = 1, p'_{n-1,1}a_n + p'_{n-1,2}a_{n-1} = 0.$$

Hence $p'_{n-1,2} = 1/a_n$ and $p'_{n-1,1} = -a_{n-1}/a_n^2$. Continuing in this manner we obtain expressions that involve powers of a_n in the denominator and products of the as. Since $a_n = 1$ the inverse contains only integers. An example of the inverse for $n = 5$ is shown below:

$$\begin{bmatrix} \frac{-a_1a_5^3+2a_2a_4a_5^2+a_3^2a_5^2-3a_3a_4^2a_5+a_4^4}{a_5^5} & \frac{-a_2a_5^2+2a_3a_4a_5-a_4^3}{a_5^4} & \frac{-a_3a_5+a_4^2}{a_5^3} & -\frac{a_4}{a_5^2} & \frac{1}{a_5} \\ \frac{-a_2a_5^2+2a_3a_4a_5-a_4^3}{a_5^4} & \frac{-a_3a_5+a_4^2}{a_5^3} & -\frac{a_4}{a_5^2} & \frac{1}{a_5} & 0 \\ \frac{-a_3a_5+a_4^2}{a_5^3} & -\frac{a_4}{a_5^2} & \frac{1}{a_5} & 0 & 0 \\ -\frac{a_4}{a_5^2} & \frac{1}{a_5} & 0 & 0 & 0 \\ \frac{1}{a_5} & 0 & 0 & 0 & 0 \end{bmatrix}$$

Problem 3.27

1.

$$L_A(b_1B_1 + b_2B_2) = A(b_1B_1 + b_2B_2) = b_1AB_1 + b_2AB_2 = b_1L_A(B_1) + b_2L_A(B_2).$$

$$R_A(b_1B_1 + b_2B_2) = (b_1B_1 + b_2B_2)A = b_1B_1A + b_2B_2A = b_1R_A(B_1) + b_2R)A(B_2).$$

$$L_AR_B(C) = L_A(CB) = ACB, R_BL_A(C) = R_B(AC) = ACB.$$

2. From the definition of the inverse of a linear map

$$L_A^{-1}L_A(B) = B.$$

Therefore

$$L_A^{-1}AB = B \Rightarrow L_A^{-1}(AB) = A^{-1}(AB) = L_{A^{-1}}(AB).$$

In the same way

$$R_A^{-1}R_A(B) = B \Rightarrow R_A^{-1}BA = (BA)A^{-1} = R_{A^{-1}}(BA).$$

If $A = PBP^{-1}$ then

$$L_A(C) = AC = PBP^{-1}C = PBL_P^{-1}(C) = PL_BL_P^{-1}(C) = L_PL_BL_P^{-1}(C)$$

and so $L_A = L_P \circ L_B \circ L_P^{-1}$. In the same way we can prove the second identity.

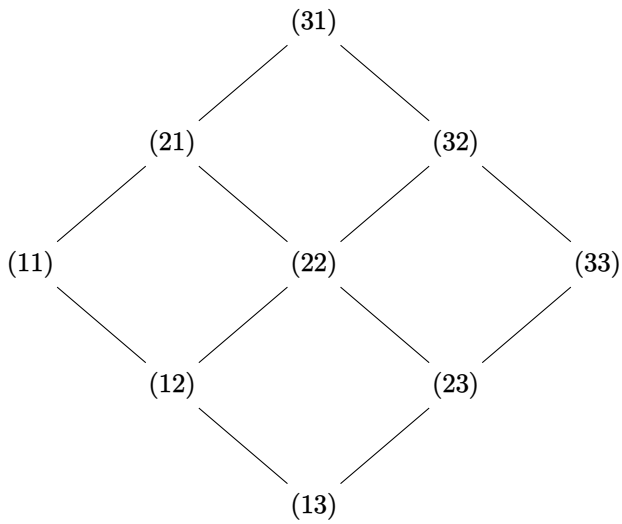
3. The ordering given is partial as it only generates an order for some of the matrices. If the first coordinate is fixed then $E_{ij} \prec E_{ik}$ for $n \geq j > k \geq 1$. From this we have the following partial orders:

$$E_{13} \prec E_{12} \prec E_{11}, \quad E_{23} \prec E_{22}, \prec E_{21} \qquad E_{33} \prec E_{32} \prec E_{31}.$$

If the second coordinate is fixed then $E_{ij} \prec E_{hj}$ for $1 \leq i < h \leq n$. We can write

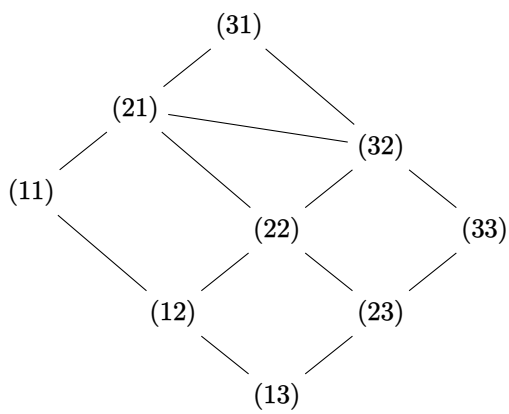
$$E_{11} \prec E_{21} \prec E_{31}, \quad E_{12} \prec E_{22} \prec E_{32}, \qquad E_{13} \prec E_{23} \prec E_{33}.$$

From these we obtain the following Hasse diagram:



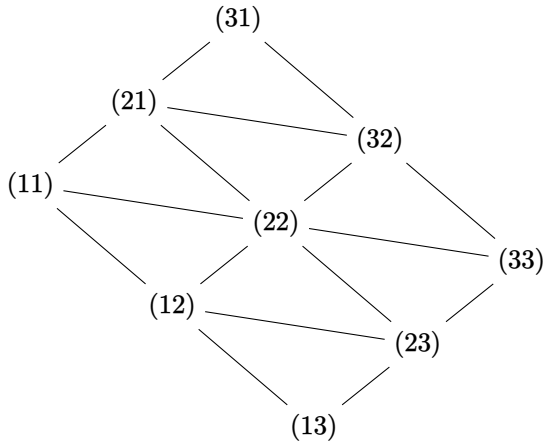
This is a partial order diagram since a number of nodes are on the same level (e.g. (21) and (32)).

There are many different total orders that we can derive from this partial order. For example suppose we decide that $E_{21} \succ E_{32}$. Then the last diagram is modified to:



Now the partial order has been extended but we still have not arrived at a total order diagram.

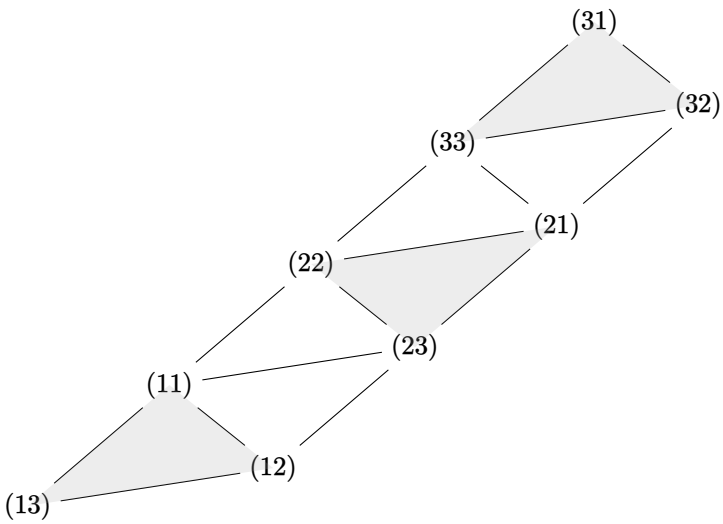
To do this lets set $E_{22} \succ E_{33}$ and $E_{12} \succ E_{23}$. Now we have



which is a total order diagram. A summary of this total order is shown in the following equation

$$(13), (23), (12), (33), (22), (11), (32), (21), (31).$$

The order given in this exercise is a total order with the following Hasse diagram:



It is easy to check that the partial order given is preserved. For any i, $E_{ij} \prec E_{ik}$ if $j > k$. These are the three triangles highlighted in the diagram above. Also for any j, $E_{ij} \prec E_{hj}$ if $h > i$ as can be easily checked.

4. We have shown that L_A is a linear map from $M_n \to M_n$. Therefore it can be uniquely determined by its action on the set of basis matrices E_{ij}. From the property $E_{ij}E_{kl} = E_{il}\delta_{jk}$ we have

$$\begin{aligned} L_A E_{ij} = AE_{ij} &= \left(\sum_{k,l} a_{kl}E_{kl}\right) E_{ij} = \sum_{k,l} a_{kl}E_{kj}\delta_{li} \\ &= \sum_k a_{ki}E_{kj}. \end{aligned}$$

Using the representation of linear maps by matrices, each column of the matrix should contain the coefficients of the E_{kj}. The complication here that each basis vector (usually) has a single index

while here the basis matrices have two indices. Hence the indices of the E matrices do not produce a straightforward order in the same way that the indices of the basis vectors $v_1, v_2, \ldots, v_m$ do. We can overcome this difficulty by using the total order given in the problem. Its (partial) explicit form is shown below:

$$E_{1n} \prec E_{1n-1} \prec \cdots \prec E_{11} \prec E_{2n} \prec E_{2n-1} \prec \cdots \prec E_{21} \cdots E_{nn} \prec E_{nn-1} \prec \cdots \prec E_{n1}.$$

Using this scheme the transformation matrix has the following structure

$$\begin{array}{c}
 \\ E_{1n} \\ E_{1n-1} \\ \vdots \\ E_{11} \\ E_{2n} \\ E_{2n-1} \\ \vdots \\ E_{21} \\ \vdots \\ E_{nn} \\ E_{nn-1} \\ \vdots \\ E_{n1}
\end{array}
\begin{array}{c}
\begin{array}{ccccccccccccc}
L_A E_{1n} & L_A E_{1n-1} & \cdots & L_A E_{11} & L_A E_{2n} & L_A E_{2n-1} & \cdots & L_A E_{21} & \cdots & L_A E_{nn} & L_A E_{nn-1} & \cdots & L_A E_{n1}
\end{array} \\
\begin{bmatrix}
a_{11} & 0 & \cdots & 0 & a_{12} & 0 & \cdots & 0 & \cdots & a_{1n} & 0 & \cdots & 0 \\
0 & a_{11} & \cdots & 0 & 0 & a_{12} & \cdots & 0 & \cdots & 0 & a_{1n} & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & a_{11} & 0 & 0 & \cdots & a_{12} & \cdots & 0 & 0 & \cdots & a_{1n} \\
a_{21} & 0 & \cdots & 0 & a_{22} & 0 & \cdots & 0 & \cdots & a_{2n} & 0 & \cdots & 0 \\
0 & a_{21} & \cdots & 0 & 0 & a_{22} & \cdots & 0 & \cdots & 0 & a_{2n} & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & a_{21} & 0 & 0 & \cdots & a_{22} & \cdots & 0 & 0 & \cdots & a_{2n} \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
a_{n1} & 0 & \cdots & 0 & a_{n2} & 0 & \cdots & 0 & \cdots & a_{nn} & 0 & \cdots & 0 \\
0 & a_{n1} & \cdots & 0 & 0 & a_{n2} & \cdots & 0 & \cdots & 0 & a_{nn} & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & a_{n1} & 0 & 0 & \cdots & a_{n2} & \cdots & 0 & 0 & \cdots & a_{nn}
\end{bmatrix}
\end{array}$$

If we interpret this structure as a "matrix-of-matrices" then each matrix element is the product of the identity matrix I with the corresponding element of A. The result is $A \otimes I$.

For R_A

$$\begin{aligned}
R_A E_{ij} = E_{ij} A = E_{ij} \left(\sum_{k,l} a_{kl} E_{kl} \right) &= \sum_{k,l} a_{kl} E_{il} \delta_{jk} \\
&= \sum_{l} a_{jl} E_{il}.
\end{aligned}$$

Now the matrix structure is different:

$$
\begin{array}{c}
 \\ E_{1n} \\ E_{1n-1} \\ \vdots \\ E_{11} \\ E_{2n} \\ E_{2n-1} \\ \vdots \\ E_{21} \\ \vdots \\ E_{nn} \\ E_{nn-1} \\ \vdots \\ E_{n1}
\end{array}
\begin{array}{c}
\begin{array}{ccccccccccccc}
L_A E_{1n} & L_A E_{1n-1} & \cdots & L_A E_{11} & L_A E_{2n} & L_A E_{2n-1} & \cdots & L_A E_{21} & \cdots & L_A E_{nn} & L_A E_{nn-1} & \cdots & L_A E_{n1}
\end{array} \\
\begin{bmatrix}
a_{nn} & a_{n-1n} & \cdots & a_{1n} & 0 & 0 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \\
a_{nn-1} & a_{n-1n-1} & \cdots & a_{1n-1} & 0 & 0 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
a_{n1} & a_{n-11} & \cdots & a_{11} & 0 & 0 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \\
0 & 0 & \cdots & 0 & a_{nn} & a_{n-1m} & \cdots & a_{1n} & \cdots & 0 & 0 & \cdots & 0 \\
0 & 0 & \cdots & 0 & a_{nn-1} & a_{n-1n-1} & \cdots & a_{1n-1} & \cdots & 0 & 0 & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & a_{21} & a_{n1} & a_{n-11} & \cdots & a_{11} & \cdots & 0 & 0 & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & a_{nn} & a_{n-1n} & \cdots & a_{1n} \\
0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & a_{nn-1} & a_{n-1n-1} & \cdots & a_{1n-1} \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & a_{n1} & a_{n-11} & \cdots & a_{11}
\end{bmatrix}
\end{array}
$$

In this case we have a "matrix-of-matrices" with only diagonal elements. These are all equal to

$$
\begin{bmatrix}
a_{nn} & a_{n-1n} & \cdots & a_{1n} \\
a_{nn-1} & a_{n-1n-1} & \cdots & a_{1n-1} \\
\vdots & \vdots & \ddots & \vdots \\
a_{n1} & a_{n-11} & \cdots & a_{11}
\end{bmatrix}
$$

The permutation matrix R has the property

$$
Rx = \begin{bmatrix} e_n^\mathsf{T} x \\ e_{n-1}^\mathsf{T} x \\ \vdots \\ e_2^\mathsf{T} x \\ e_1^\mathsf{T} x \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \\ \vdots \\ x_2 \\ x_1 \end{bmatrix}, \quad x^\mathsf{T} R = x^\mathsf{T} R^\mathsf{T} = (Rx)^\mathsf{T}.
$$

Hence,

$$
RA^\mathsf{T} = \begin{bmatrix}
a_{1n} & a_{2n} & \cdots & a_{nn} \\
a_{1n-1} & a_{2n-1} & \cdots & a_{nn-1} \\
\vdots & \vdots & \ddots & \vdots \\
a_{11} & a_{21} & \cdots & a_{n1}
\end{bmatrix}
$$

and

$$
RA^\mathsf{T} R = \begin{bmatrix}
a_{nn} & a_{n-1n} & \cdots & a_{1n} \\
a_{nn-1} & a_{n-1n-1} & \cdots & a_{1n-1} \\
\vdots & \vdots & \ddots & \vdots \\
a_{n1} & a_{n-11} & \cdots & a_{11}
\end{bmatrix}
$$

completing the derivation.

5. Using the composite index for the Kronecker product

$$(A \otimes I)_{n(i-1)+k,n(j-1)+l} = A_{ij} I_{kl} = a_{ij} \mathbb{1}_{j \geq i} \delta_{kl}.$$

If $A \otimes I$ is upper triangular we must have

$$n(j-1) + l < n(i-1) + k \Leftrightarrow (A \otimes I)_{n(i-1)+k,n(j-1)+l} = 0.$$

Since $\delta_{kl} = 0$ if $k \neq l$ the inequality is equivalent to $j < i$. If this condition holds then $\mathbb{1}_{j \geq i} = 0$ and so the proof for L_A is complete.

From the matrix expression of $RA^{\mathsf{T}}R$ it is clear that this is an upper triangular matrix. Since $I \otimes RA^{\mathsf{T}}R$ is a diagonal matrix of matrices, all elements below the diagonal are zero.

Chapter 4

A great additional reference for this chapter is Strang [1993].

Problem 4.1

1. This part is straightforward.

2. To get a formula for the inverse use the equation in (1). If $y = W_{3,3}c$, then

$$\begin{aligned}
c_1 &= \tfrac{1}{2}y_1 + \tfrac{1}{2}y_2 \\
c_2 &= + \tfrac{1}{2}y_3 + \tfrac{1}{2}y_4 \\
c_3 &= + \tfrac{1}{2}y_5 + \tfrac{1}{2}y_6 \\
c_4 &= + \tfrac{1}{2}y_7 + \tfrac{1}{2}y_8 \\
c_5 &= \tfrac{1}{2}y_1 - \tfrac{1}{2}y_2 \\
c_6 &= + \tfrac{1}{2}y_3 + \tfrac{1}{2}y_4 \\
c_7 &= + \tfrac{1}{2}y_5 - \tfrac{1}{2}y_6 \\
c_8 &= + \tfrac{1}{2}y_7 - \tfrac{1}{2}y_8
\end{aligned}$$

which translates to

$$W_{3,3}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} = \frac{1}{2}W_{3,3}^{\mathsf{T}}.$$

Next for the orthogonality of the columns of $W_{3,3}$

$$W_{3,3}^{\mathsf{T}}W_{3,3} = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix} = 2I$$

and

$$W_{3,3}W_{3,3}^{\mathsf{T}} = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix} = 2I$$

which proves that the rows and columns of $W_{3,3}^{-1}$ are orthogonal.

3. This is easier by writing each of $W_{3,1},\ W_{3,2},\ W_{3,3}$ as block 4×4 matrices

$$W_{3,1} = \begin{bmatrix} H & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}$$

$$W_{3,2} = \begin{bmatrix} E_1 & A_1 & O & O \\ E_2 & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}$$

$$W_{3,3} = \begin{bmatrix} E_1 & O & A_1 & O \\ E_2 & O & A_2 & O \\ O & E_1 & O & A_1 \\ O & E_2 & O & A_2 \end{bmatrix}$$

where $H = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, $E_1 = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right]$, $E_2 = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, $A_1 = \left[\begin{smallmatrix} 1 & 0 \\ -1 & 0 \end{smallmatrix}\right]$ and $A_2 = \left[\begin{smallmatrix} 0 & 1 \\ 0 & -1 \end{smallmatrix}\right]$. Then

$$\begin{aligned} W_{3,2}W_{3,1} &= \begin{bmatrix} E_1H & A_1 & O & O \\ E_2H & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \\ &= \begin{bmatrix} E & A_1 & O & O \\ A^{\mathsf{T}} & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \end{aligned}$$

where $E = E_1 + E_2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ and $A = A_1 + A_2 = \left[\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right]$. Finally

$$\begin{aligned} W_{3,3}W_{3,2}W_{3,1} &= \begin{bmatrix} E_1E & E_1A_1 & A_1 & O \\ E_2E & E_2A_1 & A_2 & O \\ E_1A^{\mathsf{T}} & E_1A_2 & O & A_1 \\ E_2A^{\mathsf{T}} & E_2A_2 & O & A_2 \end{bmatrix} \\ &= \begin{bmatrix} E & E_1 & A_1 & O \\ E & -E_1 & A_2 & O \\ A^{\mathsf{T}} & E_2 & O & A_1 \\ A^{\mathsf{T}} & -E_2 & O & A_2 \end{bmatrix}. \end{aligned}$$

The last matrix is equal to W_3.

4. The proof for $W_{3,1}$ is straightforward since

$$W_{3,1}^{\mathsf{T}}W_{3,1} = W_{3,1}W_{3,1}^{\mathsf{T}} = W_{3,1}^2 = \begin{bmatrix} H^2 & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} = \begin{bmatrix} 2I & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}.$$

For $W_{3,2}$ first check column orthogonality:

$$\begin{aligned} W_{3,2}^{\mathsf{T}} W_{3,2} &= \begin{bmatrix} E_1^{\mathsf{T}} & E_2^{\mathsf{T}} & O & O \\ A_1^{\mathsf{T}} & A_2^{\mathsf{T}} & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \begin{bmatrix} E_1 & A_1 & O & O \\ E_2 & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \\ &= \begin{bmatrix} E_1^{\mathsf{T}} E_1 + E_2^{\mathsf{T}} E_2 & E_1^{\mathsf{T}} A_1 + E_2^{\mathsf{T}} A_2 & O & O \\ A_1^{\mathsf{T}} E_1 + A_2^{\mathsf{T}} E_2 & A_1^{\mathsf{T}} A_1 + A_2^{\mathsf{T}} A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}. \end{aligned}$$

Use

$$E_1^{\mathsf{T}} E_1 = A_1^{\mathsf{T}} A_1 = 2E_{11},\ E_2^{\mathsf{T}} E_2 = A_2^{\mathsf{T}} A_2 = E_{22},\ E_1^{\mathsf{T}} A_1 = E_2^{\mathsf{T}} A_2 = O$$

to rewrite the last equation as

$$W_{3,2}^{\mathsf{T}} W_{3,2} = \begin{bmatrix} 2I & O & O & O \\ O & 2I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}$$

from which we conclude the columns of $W_{3,2}$ are orthogonal. For the rows

$$\begin{aligned} W_{3,2} W_{3,2}^{\mathsf{T}} &= \begin{bmatrix} E_1 & A_1 & O & O \\ E_2 & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \begin{bmatrix} E_1^{\mathsf{T}} & E_2^{\mathsf{T}} & O & O \\ A_1^{\mathsf{T}} & A_2^{\mathsf{T}} & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \\ &= \begin{bmatrix} E_1 E_1^{\mathsf{T}} + A_1 A_1^{\mathsf{T}} & E_1 E_2^{\mathsf{T}} + A_1 A_2^{\mathsf{T}} & O & O \\ E_2 A_1^{\mathsf{T}} + A_2 A_1^{\mathsf{T}} & E_2 E_2^{\mathsf{T}} + A_2 A_2^{\mathsf{T}} & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}. \end{aligned}$$

Use

$$E_1 E_1^{\mathsf{T}} = E_2 E_2^{\mathsf{T}} = E,\ A_1 A_1^{\mathsf{T}} = A_2 A_2^{\mathsf{T}} = \left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right] = L,\ E_1 E_2^{\mathsf{T}} = A_1 A_2^{\mathsf{T}} = E_2 A_1^{\mathsf{T}} = A_2 A_1^{\mathsf{T}} = O$$

to conclude that

$$W_{3,2} W_{3,2}^{\mathsf{T}} = \begin{bmatrix} 2I & O & O & O \\ O & 2I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}.$$

From these to test the orthogonality of the columns of W_3

$$\begin{aligned} W_3^{\mathsf{T}} W_3 &= W_{3,1}^{\mathsf{T}} W_{3,2}^{\mathsf{T}} W_{3,3}^{\mathsf{T}} W_{3,3} W_{3,2} W_{3,1} \\ &= W_{3,1}^{\mathsf{T}} W_{3,2}^{\mathsf{T}} \operatorname{diag}([2I, 2I, 2I, 2I]) W_{3,2} W_{3,1} \\ &= 2 W_{3,1}^{\mathsf{T}} W_{3,2}^{\mathsf{T}} W_{3,2} W_{3,1} \\ &= 2 W_{3,1}^{\mathsf{T}} \operatorname{diag}([2I, 2I, I, I]) W_{3,1} \\ &= 2 \begin{bmatrix} H & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \begin{bmatrix} 2I & O & O & O \\ O & 2I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \begin{bmatrix} H & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \\ &= 2 \begin{bmatrix} H & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} \begin{bmatrix} 2H & O & O & O \\ O & 2I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} = \begin{bmatrix} 8I & O & O & O \\ O & 4I & O & O \\ O & O & 2I & O \\ O & O & O & 2I \end{bmatrix}. \end{aligned}$$

For rows of W_3^{-1},

$$\begin{aligned} W_3^{-1}W_3^{-\mathsf{T}} &= \left(W_3^{\mathsf{T}}W_3\right)^{-1} \\ &= \begin{bmatrix} \frac{1}{8}I & O & O & O \\ O & \frac{1}{4}I & O & O \\ O & O & \frac{1}{2}I & O \\ O & O & O & \frac{1}{2}I \end{bmatrix}. \end{aligned}$$

proving orthogonality. For the rows of W_3

$$\begin{aligned} W_3W_3^{\mathsf{T}} &= W_{3,3}W_{3,2}W_{3,1}W_{3,1}^{\mathsf{T}}W_{3,2}^{\mathsf{T}}W_{3,3}^{\mathsf{T}} \\ &= W_{3,3}W_{3,2}\operatorname{diag}\left([2I, I, I, I]\right)W_{3,2}^{\mathsf{T}}W_{3,3}^{\mathsf{T}} \\ &= W_{3,3}\begin{bmatrix} E_1 & A_1 & O & O \\ E_2 & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}\begin{bmatrix} 2I & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}\begin{bmatrix} E_1^{\mathsf{T}} & E_2^{\mathsf{T}} & O & O \\ A_1^{\mathsf{T}} & A_2^{\mathsf{T}} & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}W_{3,3}^{\mathsf{T}} \\ &= W_{3,3}\begin{bmatrix} E_1 & A_1 & O & O \\ E_2 & A_2 & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}\begin{bmatrix} 2E_1^{\mathsf{T}} & 2E_2^{\mathsf{T}} & O & O \\ A_1^{\mathsf{T}} & A_2^{\mathsf{T}} & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}W_{3,3}^{\mathsf{T}} \\ &= W_{3,3}\begin{bmatrix} 2E_1E_1^{\mathsf{T}} + A_1A_1^{\mathsf{T}} & 2E_1E_2^{\mathsf{T}} + A_1A_2^{\mathsf{T}} & O & O \\ 2E_2E_1^{\mathsf{T}} + A_2A_1^{\mathsf{T}} & 2E_2E_2^{\mathsf{T}} + A_2A_2^{\mathsf{T}} & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}W_{3,3}^{\mathsf{T}} \\ &= W_{3,3}\begin{bmatrix} E+2I & O & O & O \\ O & E+2I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}W_{3,3}^{\mathsf{T}} \end{aligned}$$

The second matrix is not diagonal and so the rows of W_3 are not orthogonal. For the columns of W_3^{-1}

$$W_3^{-\mathsf{T}}W_3^{-1} = \left(W_3W_3^{\mathsf{T}}\right)^{-1}.$$

We have already shown that the matrix in the brackets is not a diagonal matrix and hence its inverse is also not a diagonal matrix. Therefore the columns of W_3^{-1} are not orthogonal.

The inverse of $W_{3,1}$ is

$$W_{3,1}^{-1} = \begin{bmatrix} H^{-1} & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix} = \begin{bmatrix} \frac{1}{2}H & O & O & O \\ O & I & O & O \\ O & O & I & O \\ O & O & O & I \end{bmatrix}.$$

To find the inverse of $W_{3,2}$ it is easier to rewrite it as a 2×2 block matrix

$$W_{3,2} = \begin{bmatrix} W_{2,2} & O \\ O & I \end{bmatrix}.$$

where

$$W_{2,2} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}.$$

From this it is clear that

$$W_{3,2}^{-1} = \begin{bmatrix} W_{2,2}^{-1} & O \\ O & I \end{bmatrix}.$$

To find the inverse of $W_{2,2}$ we follow exactly the same procedure as for $W_{3,3}$. As expected $W_{2,2}^{-1} = \frac{1}{2}W_{2,2}^{\mathsf{T}}$ and so

$$W_{3,2}^{-1} = \begin{bmatrix} \frac{1}{2}W_{2,2}^{\mathsf{T}} & O \\ O & I \end{bmatrix}.$$

Problem 4.2

1. From the scheme given for constructing the $W_{n,n}$ matrix:

$$\begin{array}{lcc}
\text{row } 1: & 1 \;\; 0 \cdots\cdots\cdots\cdots 0 & 1 \cdots\cdots\cdots\cdots\cdots\cdots 0 \\
\text{row } 2: & \underbrace{1 \;\; 0 \cdots\cdots\cdots\cdots 0}_{2^{n-1}} & \underbrace{-1 \cdots\cdots\cdots\cdots\cdots\cdots 0}_{2^{n-1}} \\
 & \vdots & \vdots \\
\text{row } 2k+1: & 0 \;\; 0 \cdots 0 \quad 1 \;\; 0 \cdots 0 & 1 \;\; 0 \cdots 0 \\
\text{row } 2k+2: & \underbrace{0 \;\; 0 \cdots 0}_{k} \quad \underbrace{1 \;\; 0 \cdots 0}_{2^{n-1}} & \underbrace{-1 \;\; 0 \cdots 0}_{2^{n-1}-k} \\
 & \vdots & \vdots
\end{array}$$

Using this template

$$\begin{aligned}
u^n(2k+1) &= u^{n-1}(k+1) + u^{n-1}(2^{n-1}+k+1) \\
u^n(2k+2) &= u^{n-1}(k+1) - u^{n-1}(2^{n-1}+k+1)
\end{aligned}$$

Using the recursive relation for the last step we get

$$\begin{aligned}
u^n(2^n) &= u^{n-1}(2^{n-1}) - u^{n-1}(2^n) \\
u^n(2^n-1) &= u^{n-1}(2^{n-1}) + u^{n-1}(2^n) \\
\vdots \qquad & \qquad \vdots \\
u^n(2^n-2k') &= u^{n-1}(2^{n-1}-k') - u^{n-1}(2^n-k') \\
u^n(2^n-2k'-1) &= u^{n-1}(2^{n-1}-k') + u^{n-1}(2^n-k') \\
\vdots \qquad & \qquad \vdots
\end{aligned}$$

If we set $k' = 2^{n-1} - k - 1$ then

$$\begin{aligned}
u^n(2^n-2k') &= u^n(2k+2) = u^{n-1}(k+1) - u^{n-1}(2^{n-1}+k+1) \\
u^n(2^n-2k'-1) &= u^n(2k+1) = u^{n-1}(k+1) + u^{n-1}(2^{n-1}+k+1)
\end{aligned}$$

so the template and the recursive relation produce identical reconstructions. The same property can be used to obtain the inverse by noting that

$$u^{n-1}(2^{n-1}-k) = \frac{1}{2}(u^n(2^n-2k-1)+u^n(2^n-2k))$$
$$u^{n-1}(2^n-k) = \frac{1}{2}(u^n(2^n-2k-1)-u^n(2^n-2k))$$

Note that now the u^n components are separated by 1. Also, since $k = 0,\ldots,2^{n-1}-1$, the formula for $u^{n-1}(2^n-k)$ applies for rows $2^{n-1}+1$ to 2^n while the formula for $u^{n-1}(2^{n-1}-k)$ applies for rows 1 to 2^{n-1}. This is the same as $W_{n,n}^{\mathsf{T}}$ except from the factor of $\frac{1}{2}$. Orthogonality of columns and rows follows from this property since

$$W_{n,n}^{\mathsf{T}}W_{n,n} = 2W_{n,n}^{-1}W_{n,n} = 2I, \quad W_{n,n}W_{n,n}^{\mathsf{T}} = 2W_{n,n}W_{n,n}^{-1} = 2I.$$

2. The formula given for $W_{n,n}$ produces a matrix with the following structure:

$$\begin{array}{c} \\ 1 \\ 1 \\ 2 \\ 2 \\ \vdots \\ 2^{n-1} \\ 2^{n-1} \end{array}
\begin{array}{c}
\begin{array}{cccccccc} 1 & 2 & \cdots & 2^{n-1} & 1 & 2 & \cdots & 2^{n-1} \end{array} \\
\begin{bmatrix}
1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\
1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\
0 & 1 & \cdots & 0 & 0 & 1 & \cdots & 0 \\
0 & 1 & \cdots & 0 & 0 & -1 & \cdots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
0 & 0 & \cdots & 1 & 0 & 0 & \cdots & 1 \\
0 & 0 & \cdots & 1 & 0 & 0 & \cdots & -1
\end{bmatrix}
\end{array}$$

This the correct expression for $W_{n,n}$. For W_n first note that from the definition of W_0, W_2

$$W_2 = \begin{bmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} & \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 1 \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} & 1 \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} W_0 \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} & I_0 \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix}$$

which is in agreement with the formula given ($I_0 = W_0 = 1$). Given the vector u^{n-1} of reconstructed coefficients then

$$\begin{aligned} W_n c &= \begin{bmatrix} W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} & I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix} \begin{pmatrix} c^1 \\ c^2 \end{pmatrix} \\ &= \left(W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) c^1 + \left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) c^2 \end{aligned}$$

where $c^1 = (c_1,\ldots,c_{2^{n-1}})$ and $c^2 = (c_{2^{n-1}+1},\ldots,c_{2^n})$. We can use the formulas given to write

$$\left(W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) c^1 = \left(W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) (c^1 \otimes 1) = (W_{n-1}c^1) \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and

$$\left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) c^2 = \left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) (c^2 \otimes 1) = c_2 \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$W_{n-1}c^1$ is the full reconstruction using c^1 which is a vector with 2^{n-1} components. c^2 is the vector of the last 2^{n-1} components of c. Then, if we abbreviate $W_{n-1}c^1$ with u_{n-1}^{n-1},

$$\left(u_{n-1}^{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)^{\mathsf{T}} = \begin{bmatrix} u_{n-1,1}^{n-1} & u_{n-1,1}^{n-1} & u_{n-1,2}^{n-1} & u_{n-1,2}^{n-1} & \cdots & u_{n-1,2^{n-1}}^{n-1} & u_{n-1,2^{n-1}}^{n-1} \end{bmatrix},$$

and

$$\left(c^2 \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)^{\mathsf{T}} = \begin{bmatrix} c_{2^{n-1}+1} & -c_{2^{n-1}+1} & c_{2^{n-1}+2} & -c_{2^{n-1}+2} & \cdots & c_{2^n} & -c_{2^n} \end{bmatrix}.$$

Combining the two equations

$$\left(u_{n-1}^{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c^2 \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)^{\mathsf{T}} =$$
$$\begin{bmatrix} u_{n-1,1}^{n-1} + c_{2^{n-1}+1} & u_{n-1,1}^{n-1} - c_{2^{n-1}+1} & u_{n-1,2}^{n-1} + c_{2^{n-1}+2} & u_{n-1,2}^{n-1} - c_{2^{n-1}+2} & \cdots & u_{n-1,2^{n-1}}^{n-1} + c_{2^n} & u_{n-1,2^{n-1}}^{n-1} - c_{2^n} \end{bmatrix}$$

which is the final equation for the reconstruction of the signal from page 107. Recall that in these equations $u^{n-1}(2^{n-1}+i) = c(2^{n-1}+i)$ since in all steps we start with $u^{j+1} = u^j$ and the only modifications apply to components $1, \ldots, 2^{j+1}$.

For the reproof, using $(A \otimes B)^{\mathsf{T}} = A^{\mathsf{T}} \otimes B^{\mathsf{T}}$,

$$\begin{aligned}
W_{n,n}W_{n,n}^{\mathsf{T}} &= \begin{bmatrix} I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} & I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix} \begin{bmatrix} I_{2^{n-1}} \otimes \begin{pmatrix} 1 & 1 \end{pmatrix} \\ I_{2^{n-1}} \otimes \begin{pmatrix} 1 & -1 \end{pmatrix} \end{bmatrix} \\
&= \left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)\left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 & 1 \end{pmatrix}\right) + \left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)\left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 & -1 \end{pmatrix}\right) \\
&= (I_{2^{n-1}}I_{2^{n-1}}) \otimes \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \end{pmatrix}\right) + (I_{2^{n-1}}I_{2^{n-1}}) \otimes \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \end{pmatrix}\right) \\
&= I_{2^{n-1}} \otimes \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + I_{2^{n-1}} \otimes \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\
&= I_{2^{n-1}} \otimes \left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\right) \\
&= I_{2^{n-1}} \otimes \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I_{2^{n-1}} \otimes I_2 = 2I_{2^n}.
\end{aligned}$$

Next we use proof by induction. From the definition of W_1

$$W_1^{\mathsf{T}}W_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = B_1.$$

Assume that $W_{n-1}^{\mathsf{T}}W_{n-1} = B_{n-1}$. Then

$$\begin{aligned}
W_n^{\mathsf{T}}W_n &= \begin{bmatrix} W_{n-1}^{\mathsf{T}} \otimes \begin{pmatrix} 1 & 1 \end{pmatrix} \\ I_{2^{n-1}} \otimes \begin{pmatrix} 1 & -1 \end{pmatrix} \end{bmatrix} \begin{bmatrix} W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} & I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix} \\
&= \begin{bmatrix} (W_{n-1}^{\mathsf{T}} \otimes \begin{pmatrix} 1 & 1 \end{pmatrix})\left(W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) & (W_{n-1}^{\mathsf{T}} \otimes \begin{pmatrix} 1 & 1 \end{pmatrix})\left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \\ (I_{2^{n-1}} \otimes \begin{pmatrix} 1 & -1 \end{pmatrix})\left(W_{n-1} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) & (I_{2^{n-1}} \otimes \begin{pmatrix} 1 & -1 \end{pmatrix})\left(I_{2^{n-1}} \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \end{bmatrix} \\
&= \begin{bmatrix} (W_{n-1}^{\mathsf{T}}W_{n-1}) \otimes \left(\begin{pmatrix} 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) & (W_{n-1}^{\mathsf{T}}I_{2^{n-1}}) \otimes \left(\begin{pmatrix} 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \\ (I_{2^{n-1}}W_{n-1}) \otimes \left(\begin{pmatrix} 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) & (I_{2^{n-1}}I_{2^{n-1}}) \otimes \left(\begin{pmatrix} 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \end{bmatrix} \\
&= \begin{bmatrix} B_{n-1} \otimes 2 & W_{n-1}^{\mathsf{T}} \otimes 0 \\ W_{n-1} \otimes 0 & I_{2^{n-1}} \otimes 2 \end{bmatrix} = 2\begin{bmatrix} B_{n-1} & O \\ O & I_{2^{n-1}} \end{bmatrix} = 2B_n.
\end{aligned}$$

3. As we have seen, the Haar components are gradually modified at each step of the reconstruction. At the ith step

$$W_{n,i}u = \begin{bmatrix} W_{i,i} & O_{2^i,2^n-2^i} \\ O_{2^n-2^i,2^i} & I_{2^n-2^i} \end{bmatrix}\begin{pmatrix} u(1:2^i) \\ u(2^i:2^n) \end{pmatrix} = \begin{pmatrix} W_{i,i}u(1:2^i) \\ I_{2^n-2^i}u(2^i:2^n) \end{pmatrix} = \begin{pmatrix} W_{i,i}u(1:2^i) \\ u(2^i:2^n) \end{pmatrix}.$$

So $W_{i,i}u(1:2^i)$ modifies the first 2^i components while $u(2^i:2^n)$ is unchanged.

To prove that

$$W_{n,n}W_{n,n-1}\cdots W_{n,1} = W_n,$$

note that for $n=1$

$$W_1 = \begin{bmatrix} W_0 \otimes e & I_1 \otimes a \end{bmatrix} = \begin{bmatrix} I_1 \otimes e & I_1 \otimes a \end{bmatrix} = W_{1,1}$$

since $W_0 = I_1 = 1$ where $e^\mathsf{T} = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $a^\mathsf{T} = \begin{pmatrix} 1 & -1 \end{pmatrix}$. If we assume that

$$W_{n-1} = W_{n-1,n-1}W_{n-1,n-2}\cdots W_{n-1,1}$$

we must show that

$$W_n = W_{n,n}W_{n,n-1}\cdots W_{n,1}.$$

From the formula for W_n

$$W_n = \begin{bmatrix} W_{n-1} \otimes e & I_{2^{n-1}} \otimes a \end{bmatrix}.$$

Write

$$(I_{2^{n-1}} \otimes e)(W_{n-1} \otimes 1) = (I_{2^{n-1}}W_{n-1}) \otimes (e \otimes 1) = W_{n-1} \otimes e.$$

Then

$$W_n = \begin{bmatrix} (I_{2^{n-1}} \otimes e)W_{n-1} & I_{2^{n-1}} \otimes a \end{bmatrix} = \begin{bmatrix} (I_{2^{n-1}} \otimes e) & I_{2^{n-1}} \otimes a \end{bmatrix} \begin{bmatrix} W_{n-1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix}.$$

$(O_{2^{n-1}} \equiv O_{2^{n-1},2^{n-1}})$. Since

$$W_{n,n} = \begin{bmatrix} (I_{2^{n-1}} \otimes e) & I_{2^{n-1}} \otimes a \end{bmatrix}$$

we can write

$$W_n = W_{n,n} \begin{bmatrix} W_{n-1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix}.$$

Using $W_{n-1} = W_{n-1,n-1}W_{n-1,n-2}\cdots W_{n-1,1}$ and the property of diagonal matrices

$$\begin{aligned} \begin{bmatrix} W_{n-1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix} &= \begin{bmatrix} W_{n-1,n-1}W_{n-1,n-2}\cdots W_{n-1,1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix} \\ &= \begin{bmatrix} W_{n-1,n-1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix} \begin{bmatrix} W_{n-1,n-2} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix} \cdots \begin{bmatrix} W_{n-1,1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix}. \end{aligned}$$

From the formula for $W_{n,i}$

$$W_{n,n-1} = \begin{bmatrix} W_{n-1,n-1} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix}.$$

Next since

$$W_{n-1,n-2} = \begin{bmatrix} W_{n-2,n-2} & O_{2^{n-2}} \\ O_{2^{n-2}} & I_{2^{n-2}} \end{bmatrix}$$

we have

$$\begin{bmatrix} W_{n-1,n-2} & O_{2^{n-1}} \\ O_{2^{n-1}} & I_{2^{n-1}} \end{bmatrix} = \begin{bmatrix} W_{n-2,n-2} & O_{2^{n-2}} & O_{2^{n-2}} & O_{2^{n-2}} \\ O_{2^{n-2}} & I_{2^{n-2}} & O_{2^{n-2}} & O_{2^{n-2}} \\ O_{2^{n-2}} & O_{2^{n-1}} & I_{2^{n-2}} & O_{2^{n-2}} \\ O_{2^{n-2}} & O_{2^{n-1}} & O_{2^{n-2}} & I_{2^{n-2}} \end{bmatrix}.$$

This matrix is

$$W_{n,n-2} = \begin{bmatrix} W_{n-2,n-2} & O_{2^{n-2},2^n-2^{n-2}} \\ O_{2^n-2^{n-2},2^{n-2}} & I_{2^n-2^{n-2}} \end{bmatrix}.$$

We can show the same for all other diagonal matrices and this completes the proof.

To prove that the columns of $W_{n,k}$ are orthogonal write

$$W_{n,k}^{\mathsf{T}} W_{n,k} = \begin{bmatrix} W_{k,k}^{\mathsf{T}} & O \\ O & I \end{bmatrix} \begin{bmatrix} W_{k,k} & O \\ O & I \end{bmatrix} = \begin{bmatrix} W_{k,k}^{\mathsf{T}} W_{k,k} & O \\ O & I \end{bmatrix}$$

Next

$$\begin{aligned} W_{k,k}^{\mathsf{T}} W_{k,k} &= \begin{bmatrix} I \otimes \begin{pmatrix} 1 & 1 \end{pmatrix} \\ I \otimes \begin{pmatrix} 1 & -1 \end{pmatrix} \end{bmatrix} \begin{bmatrix} I \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} & I \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{bmatrix} \\ &= \begin{bmatrix} \left(I \otimes \begin{pmatrix} 1 & 1 \end{pmatrix}\right)\left(I \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) & \left(I \otimes \begin{pmatrix} 1 & 1 \end{pmatrix}\right)\left(I \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \\ \left(I \otimes \begin{pmatrix} 1 & -1 \end{pmatrix}\right)\left(I \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) & \left(I \otimes \begin{pmatrix} 1 & -1 \end{pmatrix}\right)\left(I \otimes \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) \end{bmatrix} \\ &= \begin{bmatrix} I \otimes 2 & I \otimes 0 \\ I \otimes 0 & I \otimes 2 \end{bmatrix} = 2I. \end{aligned}$$

Since $W_{k,k}^{\mathsf{T}} W_{k,k}$ is diagonal, $W_{n,k}^{\mathsf{T}} W_{n,k}$ is diagonal and this completes the proof. For the rows of $W_{n,k}$

$$W_{n,k} W_{n,k}^{\mathsf{T}} = \begin{bmatrix} W_{k,k} & O \\ O & I \end{bmatrix} \begin{bmatrix} W_{k,k}^{\mathsf{T}} & O \\ O & I \end{bmatrix} = \begin{bmatrix} W_{k,k} W_{k,k}^{\mathsf{T}} & O \\ O & I \end{bmatrix}.$$

We have shown above that $W_{k,k} W_{k,k}^{\mathsf{T}}$ is diagonal and so conclude that the rows of $W_{n,k}$ are orthogonal.

For the rows of W_n

$$\begin{aligned} W_n W_n^{\mathsf{T}} &= \begin{bmatrix} W_{n-1} \otimes e & I \otimes a \end{bmatrix} \begin{bmatrix} W_{n-1}^{\mathsf{T}} \otimes e^{\mathsf{T}} \\ I \otimes a^{\mathsf{T}} \end{bmatrix} \\ &= (W_{n-1} \otimes e)(W_{n-1}^{\mathsf{T}} \otimes e^{\mathsf{T}}) + (I \otimes a)(I \otimes a^{\mathsf{T}}) \\ &= (W_{n-1} W_{n-1}^{\mathsf{T}}) \otimes (ee^{\mathsf{T}}) + I \otimes (aa^{\mathsf{T}}). \end{aligned}$$

We have a recursive relation which only contains matrices like $E = ee^{\mathsf{T}}$ and $A = aa^T$ which are not diagonal. If we expand the expression we obtain powers of E and A which are also not diagonal. So the rows of W_n are not orthogonal. For the columns of W_n^{-1}

$$W_n^{-\mathsf{T}} W_n - 1 = (W_n W_n^{\mathsf{T}})$$

and we reach the same conclusion. For the inverse of $W_{n,k}$, since it is a diagonal block matrix

$$W_{n,k}^{-1} = \begin{bmatrix} W_{k,k}^{-1} & O \\ O & I \end{bmatrix} = \begin{bmatrix} \frac{1}{2} W_{k,k}^{\mathsf{T}} & O \\ O & I \end{bmatrix}$$

as we have shown in (1).

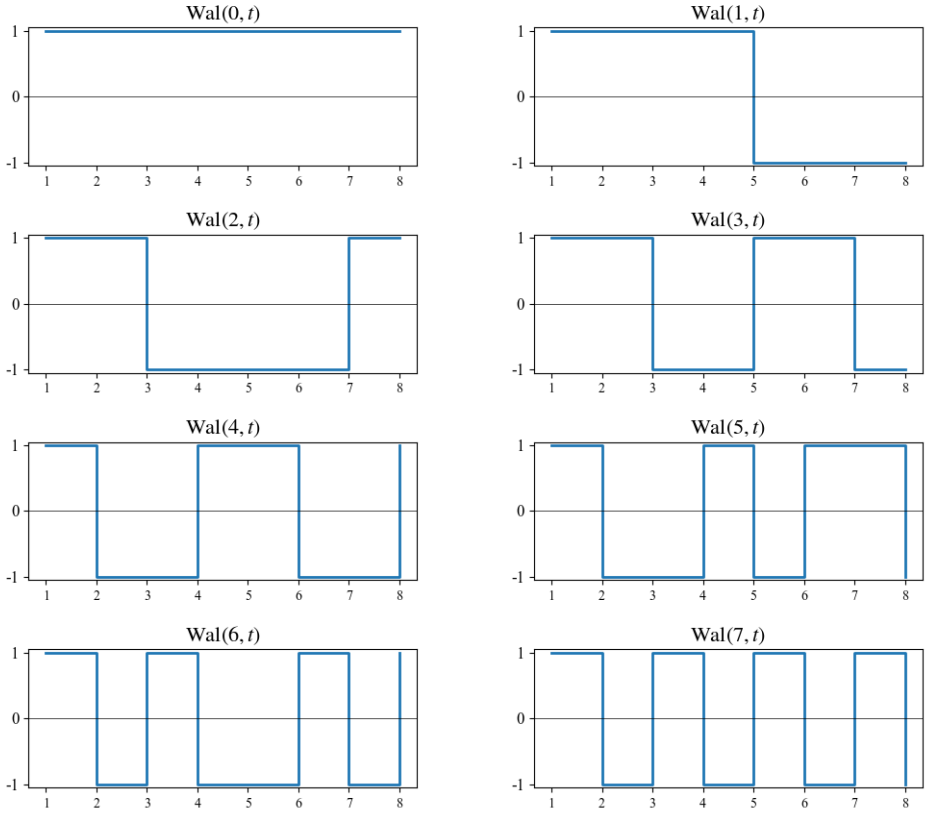
Problem 4.3

H is a Hadamard matrix with the property $H^{\mathsf{T}} H = nI_n$. The matrix given is $H_2 \otimes H$; therefore its elements are also ± 1. We have

$$(H_2 \otimes H)^{\mathsf{T}} (H_2 \otimes H) = (H_2^{\mathsf{T}} \otimes H^{\mathsf{T}})(H_2 \otimes H) = (H_2^{\mathsf{T}} H_2) \otimes (H^{\mathsf{T}} H) = 2I_2 \otimes nI_n = 2nI_2 \otimes I_n = 2nI_{2n}$$

completing the proof.

Problem 4.4



Problem 4.5

For any Hadamard matrix

$$H_{2^n} = \begin{bmatrix} H_{2^{n-1}} & H_{2^{n-1}} \\ H_{2^{n-1}} & -H_{2^{n-1}} \end{bmatrix}.$$

If we split a $2^n \times 1$ vector into two vectors $x^1_{2^{n-1}}$, $x^2_{2^{n-1}}$ it is easy to show that

$$H_{2^n} x = \begin{bmatrix} H_{2^{n-1}} x^1_{2^{n-1}} + H_{2^{n-1}} x^2_{2^{n-1}} \\ H_{2^{n-1}} x^1_{2^{n-1}} - H_{2^{n-1}} x^2_{2^{n-1}} \end{bmatrix}.$$

In turn, for $H_{2^{n-1}} x_1$ we can repeat the same procedure. We do this until we reach the simple evaluation for H_2. As shown in the algorithm below, for each of the $n/2$ times we repeat this procedure, we have two recursive calls, so a total of n recursive calls.

Algorithm 1 Recursive Hadamard Transform

Require: $x \in \mathbb{R}^{2^n}$, integer $n \geq 1$
Ensure: $y = H_{2^n} x$

```
function HADAMARD(x, n)
    if n = 1 then
        return [x_1 + x_2   x_1 - x_2]
    end if
    Split x into x^1, x^2 ∈ R^{2^{n-1}}
    y^1 ← HADAMARD(x^1, n - 1)
    y^2 ← HADAMARD(x^2, n - 1)
    return [y^1 + y^2   y^1 - y^2]
end function
```

Chapter 5

Problem 5.1

Pick two vectors $v \in V$ and $w \in W$; then $v+w \in V \cup W$. There are three cases: first, $v+w \in (V \cup W) \setminus W$ which contradicts the fact that V is a subspace unless $W \subseteq V$. Second, $v+w \in (V \cup W) \setminus V$ which contradicts the fact that W is a subspace unless $V \subseteq W$. Finally $v+w \in V \cap W$ which again contradicts the subspace properties of V, W unless $V = W$. We conclude that we either have $V \subseteq W$ or $W \subseteq V$.

Problem 5.2

First pick x such that $f(x) = 0$. If $x \in \operatorname{im} f$ then $x = f(y)$ for some $y \in E$. From the definition of f, $f(x) = f \circ f(y) = f(y)$ so $y \in \ker f$. Therefore $f(y) = 0$ which means that only if $x = 0$ then $x \in \ker f$, $\operatorname{im} f$. Hence $\ker f \cap \operatorname{im} f = \{0\}$ and E is the direct sum of these two subspaces.

Problem 5.3

1. From the definition of $\ker a$, $(u_1, \ldots, u_p) \in \ker a$ if

$$u_1 + \cdots + u_i + \cdots + u_p = 0.$$

This is equivalent to $u_i = -\sum_{j \neq i} u_j$ for all $i = 1, \ldots, p$. This can only hold if

$$u_i \in U_i \cap \sum_{j \neq i} U_j, \quad i = 1, \ldots, p.$$

Hence we obtain sets $Z_1, \ldots, Z_p$.

If we require that $U_i \cap \sum_{j \neq i} U_j = \{0\}$ it does not follow that $Z_i = \{0\}$. For the other subspaces we could have $U_j \cap \sum_{k \neq j} U_k \neq \{0\}$ and this would produce nonzero elements in Z_i. If we require that $U_i \cap \sum_{j \neq i} U_j = \{0\}$ for all $i = 1, \ldots, p$ then $Z_i = \{0\}$ for all $i = 1, \ldots, p$ and $\ker a = \{0\}$.
2. If the condition given holds then $\ker a = \{0\}$ and a is injective. When map a is injective then it is a linear isomorphism between $U_1 \times \cdots \times U_p$ and $U_1 \oplus \cdots \oplus U_p$. Hence $U_1 + \cdots + U_p$ is a direct sum. Conversely, if $U_1 + \cdots + U_p$ is a direct sum then a is an injective linear map and $\ker a = \{0\}$. From the definition of $\ker a$ this can hold only if $U_i \cap \sum_{j \neq i} U_j = \{0\}$ for all $i = 1, \ldots, p$. This completes the proof.

Problem 5.4

1. From the definition, for some vector $x \in E$

$$\mathrm{id}_E\, x = x = f_1 x + \cdots + f_p x.$$

Therefore $x \in \sum_i \operatorname{im} f_i$ and since $\operatorname{im} \mathrm{id}_E = E$ we must also have $E = \sum_i \operatorname{im} f_i$. If we assume that $f_i \circ f_j = 0$ then $\operatorname{im} f_i \cap \operatorname{im} f_j = \{0\}$ if $i \neq j$ and so $E = \operatorname{im} f_1 \oplus \cdots \oplus \operatorname{im} f_p$. Therefore f_i partitions

E and so it must be a projection with the property $f_i \circ f_i = f_i$. Conversely, note that for any linear map f the trace of its matrix representation is basis independent. We have already seen that given a basis transformation any matrix F has the property

$$F' = PFP^{-1}$$

where P is the change of basis matrix. From the cyclical property of the trace operator

$$\operatorname{tr}(F') = \operatorname{tr}\big(PFP^{-1}\big) = \operatorname{tr}\big(P^{-1}PF\big) = \operatorname{tr}(F)$$

and so the trace of a linear map is invariant. Since

$$\mathrm{id}_E = f_1 + \cdots + f_p$$

it follows that

$$\dim E = \operatorname{tr}(\mathrm{id}_E) = \operatorname{tr}(f_1) + \cdots + \operatorname{tr}(f_p).$$

Now under some basis the matrix of a projection f has the form $\begin{bmatrix} I_r & O \\ O & O \end{bmatrix}$ where r is the dimension of $\operatorname{im} f$. $\operatorname{tr} f = \dim \operatorname{im} f$ and so we must have

$$\dim E = \dim \operatorname{im} f_1 + \cdots + \dim \operatorname{im} f_p.$$

This only holds if $\operatorname{im} f_i \cap \operatorname{im} f_j = \{0\}$ for any $i \neq j$. To see this note that

$$\dim \mathrm{id}_E = \dim \operatorname{im} f_1 + \dim \operatorname{im}(f_2 + \cdots f_p) - \dim \operatorname{im} f_1 \cap \operatorname{im}(f_2 + \cdots f_p).$$

If we write the same for $\dim \operatorname{im}(f_2 + \cdots f_p)$ until it is fully resolved we obtain

$$\dim \mathrm{id}_E = \dim \operatorname{im} f_1 + \cdots + \dim \operatorname{im} f_p + \underbrace{\cdots\cdots\cdots\cdots}_{\text{lots of intersections}}.$$

Only if all intersections have 0 as their common element does equality hold.

Define the projection operator $\pi_i : E \to U_i$. This, obviously, has the property $\pi_i \circ \pi_i = \pi_i$. Next, from the property given

$$\begin{aligned}(\pi_1 + \cdots + \pi_n) \circ (\pi_1 + \cdots + \pi_n) &= \pi_n \circ \pi_n + \pi_n \circ (\pi_1 + \cdots + \pi_{n-1}) + (\pi_1 + \cdots + \pi_{n-1}) \circ \pi_n \\ &\quad + (\pi_1 + \cdots + \pi_{n-1}) \circ (\pi_1 + \cdots + \pi_{n-1}) \\ &= \pi_n + (\pi_1 + \cdots + \pi_{n-1}) \circ (\pi_1 + \cdots + \pi_{n-1}).\end{aligned}$$

Repeating the same procedure for all partial sums we obtain

$$(\pi_1 + \cdots + \pi_n) \circ (\pi_1 + \cdots + \pi_n) = \pi_1 + \cdots + \pi_n$$

and so $\pi = \pi_1 + \cdots + \pi_n$ is a projection. If we replace id_E (which is a projection) with π and E with $U_1 + \cdots + U_p$ in the first part of this question we obtain the same results. Hence we conclude that $\pi_i \circ \pi_j = 0$ for $i \neq j$ and so $U_1 + \cdots + U_p$ is a direct sum. Conversely, if $U_1 + \cdots + U_p$ is a direct sum then $\pi_i \circ \pi_j = 0$ for $i \neq j$ and the property given holds.

Problem 5.5

1. An involution is invertible since if $f(x) = 0$ we also have $f(f(x)) = x$. Since f is a linear map $x = 0$ and so $\ker f = \{0\}$. Furthermore, $f \circ f = \mathrm{id} \Rightarrow f^{-1} = f$.
2. f has the property

$$f\left(\frac{u + f(u)}{2}\right) = \frac{f(u) + f(f(u))}{2} = u, \quad u \in E_1$$
$$f\left(\frac{u + f(u)}{2}\right) = \frac{f(u) + f(f(u))}{2} = 0, \quad u \in E_{-1}$$

and

$$f\left(\frac{u-f(u)}{2}\right)=\frac{f(u)-f(f(u))}{2}=0,\quad u\in E_1$$
$$f\left(\frac{u-f(u)}{2}\right)=\frac{f(u)-f(f(u))}{2}=-u,\quad u\in E_{-1}$$

Therefore $\pi_1 = f\circ\frac{1}{2}(f+\mathrm{id})$ is a projection $E\to E_1$ while $\pi_{-1}=f\circ\frac{1}{2}(f-\mathrm{id})$ is a projection $E\to E_{-1}$ ($\ker\pi_1=E_{-1}$ and $\ker\pi_{-1}=E_1$). Since for any $u\in E$

$$u=\frac{u+f(u)}{2}+\frac{u-f(u)}{2}$$

we must have

$$f(E)=E=\operatorname{im}\pi_1+\operatorname{im}\pi_{-1}$$

since for f to be invertible it must be bijective. Combining this with the kernels of π_1, π_{-1} we conclude that $E=E_1\oplus E_{-1}$.

3. Since f is an involution, if F is the matrix representation of f in some basis, then $F^2=I_n$. Write F in block form

$$F=\begin{bmatrix}A & B\\ C & D\end{bmatrix}.$$

Since $\pi_1=f\circ\frac{1}{2}(f+\mathrm{id})$, we have

$$\begin{bmatrix}A & B\\ C & D\end{bmatrix}\frac{1}{2}\left(\begin{bmatrix}A & B\\ C & D\end{bmatrix}+\begin{bmatrix}I_k & O\\ O & I_{n-k}\end{bmatrix}\right)=\frac{1}{2}\left(\begin{bmatrix}I_k & O\\ O & I_{n-k}\end{bmatrix}+\begin{bmatrix}A & B\\ C & D\end{bmatrix}\right)=\begin{bmatrix}I_k & O\\ O & O\end{bmatrix}.$$

The matrix on the right-hand side is the matrix representation of π_1 in the same basis. From the above equation we have $A^2+AB=A$ and $CA+CD=C$. Since $F^2=I_n$ it is easy to see that $B=C=O$, $A=I_k$ and $D=I_{n-k}$. We can check that

$$\begin{bmatrix}I_k & O\\ O & -I_{n-k}\end{bmatrix}\begin{bmatrix}I_k & O\\ O & -I_{n-k}\end{bmatrix}=\begin{bmatrix}I_k & O\\ O & I_{n-k}\end{bmatrix}$$

and

$$\frac{1}{2}\left(\begin{bmatrix}I_k & O\\ O & I_{n-k}\end{bmatrix}-\begin{bmatrix}I_k & O\\ O & -I_{n-k}\end{bmatrix}\right)=\begin{bmatrix}O & O\\ O & I_{n-k}\end{bmatrix}.$$

the matrix for π_{-1}. It is easier to give a geometric interpretation of the action of f if we consider matrix F in $\mathbb{R}^2$. If under some basis F is represented by $\left[\begin{smallmatrix}1 & 0\\ 0 & -1\end{smallmatrix}\right]$ then f is a reflection across the x-axis. For vectors like $\vec{x}=\begin{pmatrix}x & 0\end{pmatrix}$, $F\vec{x}=\vec{x}$ while for vectors like $\vec{y}=\begin{pmatrix}0 & y\end{pmatrix}$, $F\vec{y}=-\vec{y}$. Any other vector $\begin{pmatrix}x & y\end{pmatrix}$ is also reflected across the x-axis. If we extend the same idea to $\mathbb{R}^n$ we can see that f produces the reflection of any vector across the subspace E_1.

Problem 5.6

An upper Hessenberg matrix looks something like this

$$H=\begin{array}{c}\\ j{=}1\\ j{=}2\\ j{=}3\\ \vdots\\ j{=}n\end{array}\begin{array}{c}\begin{array}{ccccc}k{=}1 & k{=}2 & k{=}3 & \cdots & k{=}n\end{array}\\ \begin{bmatrix}0 & * & * & \cdots & *\\ 0 & 0 & * & \cdots & *\\ 0 & 0 & 0 & \cdots & *\\ \vdots & \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & 0 & \cdots & 0\end{bmatrix}\end{array},$$

while an unreduced upper Hessenberg matrix looks like this

$$H_u = \begin{array}{c} \\ j{=}1 \\ j{=}2 \\ j{=}3 \\ \vdots \\ j{=}n{-}1 \\ j{=}n \end{array} \begin{array}{c} \begin{array}{cccccc} k{=}1 & k{=}2 & k{=}3 & \cdots & k{=}n{-}1 & k{=}n \end{array} \\ \begin{bmatrix} 0 & * & * & \cdots & * & * \\ * & 0 & * & \cdots & * & * \\ 0 & * & 0 & \cdots & * & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & * \\ 0 & 0 & 0 & \cdots & * & 0 \end{bmatrix} \end{array}.$$

Consider a vector x such that $Hx = 0$. We obtain the following system of equations

$$\begin{aligned} 0 \cdot x_1 + h_{12}x_2 + h_{13}x_3 + \cdots + h_{1n-2}x_{n-3} + h_{1n-1}x_{n-1} + h_{1n}x_n &= 0 \\ h_{21} \cdot x_1 + 0 \cdot x_2 + h_{23}x_3 + \cdots + h_{2n-2}x_{n-2} + h_{2n-1}x_{n-1} + h_{2n}x_n &= 0 \\ &\vdots \\ 0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 + \cdots + h_{n-1,n-2} \cdot x_{n-2} + 0 \cdot x_{n-1} + h_{n-1,n}x_n &= 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 + \cdots + 0 \cdot x_{n-2} + h_{n,n-1}x_{n-1} + 0 \cdot x_n &= 0 \end{aligned}$$

From the last equation we have $x_{n-1} = 0$. From the second to last equation we have

$$x_{n-2} = -\frac{h_{n-1,n}}{h_{n-1,n-2}} x_n.$$

From the third to last equation we have

$$x_{n-3} = -\frac{h_{n-2,n}}{h_{n-2,n-3}} x_n.$$

etc. Continuing in this way we can express all x_i as a function of x_n. Hence the null space of H is one-dimensional and so $\dim(\ker(H)) = 1$.

Problem 5.7

1. If $Ax = 0$ then $A^\mathsf{T}Ax = 0$. Now suppose that for some vector y, $Ay \neq 0$ is in the nullspace of A^T. Then we can write

$$y^\mathsf{T}A^\mathsf{T}Ay = 0 \Rightarrow \|Ay\|^2 = 0.$$

This is only possible if $Ay = 0$ which contradicts the assumption. Hence $\ker(A) = \ker(A^\mathsf{T}A)$. Since by the Rank-Nullity Theorem

$$k = \operatorname{rank}(A^\mathsf{T}A) + \dim\ker(A^\mathsf{T}A)$$

and

$$k = \operatorname{rank}(A) + \dim\ker(A)$$

use of $\ker(A^\mathsf{T}A) = \ker(A)$ completes the proof. Exactly the same argument can be used to show that $\operatorname{rank}(AA^\mathsf{T}) = \operatorname{rank}(A)$.

2. Since the columns of A are linearly independent vectors the only vector x for which $Ax = 0$ is the zero vector. Hence $\ker(A) = \{0\}$ and so $\ker(A^\mathsf{T}A) = \{0\}$. From the Rank-Nullity Theorem we have $\operatorname{rank}(A^\mathsf{T}A) = k$. It is invertible since it is surjective (its columns span a k dimensional space) and injective. For P,

$$P^2 = A(A^\mathsf{T}A)^{-1}A^TA(A^\mathsf{T}A)^{-1}A^\mathsf{T} = A(A^\mathsf{T}A)^{-1}(A^\mathsf{T}A)(A^\mathsf{T}A)^{-1}A^\mathsf{T} = A(A^\mathsf{T}A)^{-1}A^\mathsf{T} = P$$

and, using $(ABC)^\mathsf{T} = C^\mathsf{T}B^\mathsf{T}A^\mathsf{T}$

$$P^\mathsf{T} = A(A^\mathsf{T}A)^{-\mathsf{T}}A^\mathsf{T} = A(A^\mathsf{T}A)^{-1}A^\mathsf{T} = P.$$

When $k = 1$, A is a column vector and

$$P = \|A\|^{-2} A A^{\mathsf{T}}$$

is a rank-1 projection matrix that maps any vector onto the line spanned by A.

3. We note again that since

$$A = \begin{bmatrix} a_1 & \cdots & a_k \end{bmatrix}$$

then

$$Ax = x_1 a_1 + \cdots + x_k a_k.$$

and so $\operatorname{im} A = \operatorname{span}(a_1, \ldots, a_k)$. Now

$$Px = A(A^{\mathsf{T}} A)^{-1} A^{\mathsf{T}} x = A((A^{\mathsf{T}} A)^{-1} A^{\mathsf{T}} x)$$

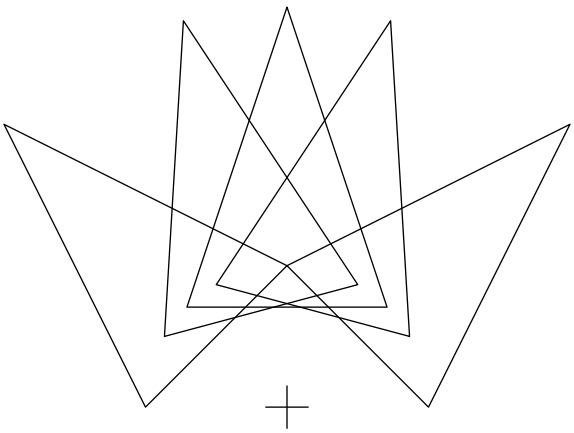
and so $\operatorname{im} P \subseteq \operatorname{im} A$. Next if $x = Ay$ then

$$PAy = A(A^{\mathsf{T}} A)^{-1} A^{\mathsf{T}} A y = Ay$$

and since P is a projection, $\operatorname{im} A \subseteq \operatorname{im} P$. Hence $\operatorname{im} P = \operatorname{im} A$. For the nullspace of P it is clear that if $A^{\mathsf{T}} x = 0$ then $Px = 0$. The nullspace of A is $\{0\}$ and so any vector that $(A^{\mathsf{T}} A)^{-1}$ maps to zero will be part of the nullspace of P. Since $A^{\mathsf{T}} A$ is invertible the only vector with this property is the zero vector. Hence $\ker P = \ker A^{\mathsf{T}}$. We note that a vector x with the property $A^{\mathsf{T}} x = 0$ is orthogonal to all columns of A and so $\ker A^{\mathsf{T}} = (\operatorname{im} A)^{\perp}$. Since $\operatorname{im} P = \operatorname{im} A$, $\ker P = (\operatorname{im} P)^{\perp}$ and so $\mathbb{R}^n = \ker P \oplus \operatorname{im} P$.

Problem 5.8

1.



2.

$$\begin{Bmatrix} x\cos\theta - y\sin\theta = x' \\ x\sin\theta + y\cos\theta = y' \end{Bmatrix} \Leftrightarrow \begin{Bmatrix} x'\cos\theta + y'\sin\theta = x \\ -x'\sin\theta + y'\cos\theta = y \end{Bmatrix}$$

Therefore

$$R_\theta^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} = R_{-\theta}.$$

3.

$$\begin{aligned} R_\theta R_\phi &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta\cos\phi - \sin\theta\sin\phi & -\cos\theta\sin\phi - \sin\theta\cos\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & -\sin\theta\sin\phi + \cos\theta\cos\phi \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi) \end{bmatrix} = R_{\theta+\phi} = R_{\phi+\theta} = R_\phi R_\theta.$$

Denote the set of plane rotation matrices by $\mathcal{R}$. This set equipped with the binary operation of matrix multiplication is a commutative group if it satisfies the following properties:

1. Commutative law: $R_\theta R_\phi = R_\phi R_\theta$ as shown above.
2. Associative law: $R_\theta(R_\phi R_\psi) = (R_\theta R_\phi)R_\psi$ since $\theta + (\phi + \psi) = (\theta + \phi) + \psi$.
3. For all $R_\theta \in \mathcal{R}$, $R_0 R_\theta = R_\theta R_0 = R_\theta$ (R_0 is the identity element).
4. For all $R_\theta \in \mathcal{R}$, $R_\theta R_{-\theta} = R_{-\theta} R_\theta = R_0$ (existence of inverse).

Problem 5.9

1. $c^\mathsf{T} = \begin{pmatrix} c_1 & c_2 \end{pmatrix}$ is a unique fixed point if

$$c = R_\theta c + a \Leftrightarrow (I - R_\theta)c = a.$$

$I - R_\theta$ is invertible since

$$\begin{bmatrix} 1-\cos\theta & -\sin\theta \\ \sin\theta & 1-\cos\theta \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \theta \in (0, 2\pi).$$

This is easy to prove since the rows of $I - R_\theta$ are orthogonal vectors and, if a third vector $\begin{pmatrix} x & y \end{pmatrix}$ is orthogonal to these two vectors, the vector space must have dimension 3 which does not hold. To find the inverse make the assumption that

$$(I - R_\theta)\lambda R_\phi = I$$

where λ is a scalar. Then we must have

$$\lambda(R_\phi - R_{\phi+\theta}) = I.$$

Since

$$R_\psi + R_{-\psi} = 2\cos\psi I$$

and

$$-R_\psi = R_{\pm\pi} R_\psi = R_{\pm\pi+\psi}$$

we must have

$$\phi = -(-\pi + \phi + \theta) \Rightarrow \phi = \frac{1}{2}(\pi - \theta)$$

and $\lambda = 1/(2\cos\left(\frac{1}{2}(\pi-\theta)\right)) = 1/(2\sin\left(\frac{1}{2}\theta\right))$.

2. It is clear that if

$$y = R_\theta x + a$$

then

$$y - c = R_\theta x + a - c.$$

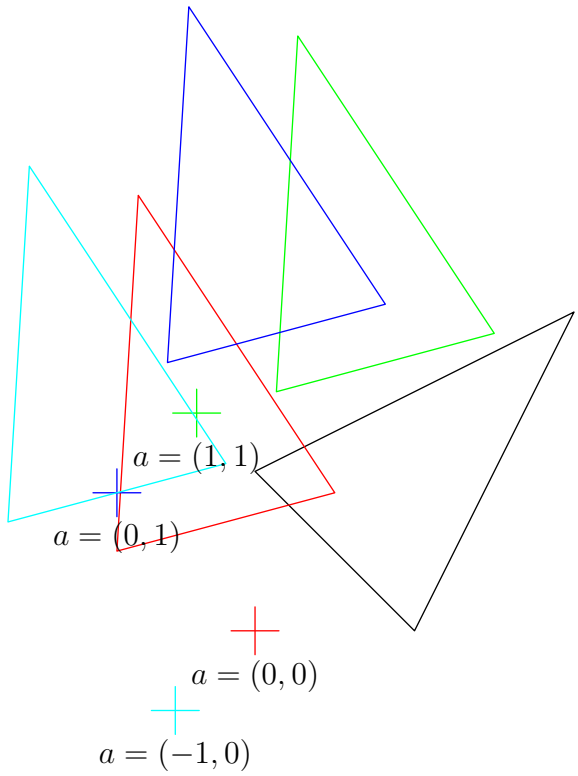
Next

$$a - c = (I - R_\theta)c - Ic = -R_\theta c.$$

Hence

$$y - c = R_\theta(x - c).$$

3.



If $\theta = 2k\pi$ then $y = x + a$ so we have a translation.

4. $R_{\theta,(a_1,a_2)}$ corresponds to the following transformations

$$x \xrightarrow{T} x - c,\ x - c \xrightarrow{R_\theta} y - c,\ y - c \xrightarrow{T} y$$

where T (resp. R) indicates a translation (resp. rotation). Given c the reverse procedure is

$$y \xrightarrow{T} y - c,\ y - c \xrightarrow{R_{-\theta}} x - c,\ x - c \xrightarrow{T} x.$$

This suggests that the inverse is of the form $R_{-\theta,(a_1,a_2)}$ since both transformations use c. However given that in the reverse procedure the rotation is by $-\theta$, (a_1, a_2) cannot produce the same c. So instead the inverse must be of the form $R_{-\theta,(b_1,b_2)}$. Since both must produce the same c,

$$\frac{1}{2\sin\frac{\theta}{2}} R_{\frac{\pi}{2}-\frac{\theta}{2}} a = -\frac{1}{2\sin\frac{\theta}{2}} R_{\frac{\pi}{2}+\frac{\theta}{2}} b,$$

or

$$R_{\frac{\pi}{2}-\frac{\theta}{2}} a = -R_{\frac{\pi}{2}+\frac{\theta}{2}} b.$$

Use $R_{-\pi} = -1$ to get

$$R_{\frac{\pi}{2}-\frac{\theta}{2}} a = R_{-\frac{\pi}{2}+\frac{\theta}{2}} b.$$

Therefore

$$b = R^2_{\frac{\pi}{2}-\frac{\theta}{2}} a = R_{\pi-\theta} a.$$

5.

$$R_{\alpha,a} x = R_\alpha x + a$$

and

$$R_{\beta,b} x = R_\beta x + b.$$

Therefore

$$R_{\beta,b} \circ R_{\alpha,a}x = R_\beta R_\alpha x + R_\beta a + b$$

and so

$$R_{\beta,b} \circ R_{\alpha,a} = R_{\alpha+\beta,t}$$

where $t = R_\beta a + b$.

In general

$$R_{\beta,b} \circ R_\alpha x = R_{\alpha+\beta}x + b$$

while

$$R_\alpha \circ R_{\beta,b}x = R_{\alpha+\beta}x + R_\alpha b$$

so the two compositions are different.

A nonabelian group is a set equiped with a binary operation for which

1. The associative law holds. Use (5) to write

$$R_{\alpha,a} \circ (R_{\beta,b} \circ R_{\gamma,c}) = R_{\alpha,a} \circ R_{\beta+\gamma,R_\beta c+b} = R_{\alpha+\beta+\gamma,R_\alpha(R_\beta c+b)+a}$$

   Also

$$(R_{\alpha,a} \circ R_{\beta,b}) \circ R_{\gamma,c} = R_{\alpha+\beta,R_\alpha b+a} \circ R_{\gamma,c} = R_{\alpha+\beta+\gamma,R_{\alpha+\beta}c+R_\alpha b+a}.$$

   It is easy to check that the two expressions are equal.

2. The identity element is $R_{0,0}$ which is an affine map.

3. In (4) we have shown the existence of an inverse for every affine map.

To prove that $R_{\beta,b} \circ R_{\alpha,a}$ is not a translation

$$R_{\beta,b} \circ R_{\alpha,a} = R_{\alpha+\beta,R_\beta a+b}.$$

This is a translation only if $\alpha + \beta = 2\pi k$ for all $k \in \mathbb{Z}$. If $a = 0$ its center of rotation is

$$c = \frac{1}{2\sin\frac{\alpha+\beta}{2}} R_{\frac{\pi}{2}-\frac{\alpha+\beta}{2}} b.$$

If $\alpha + \beta = 2\pi k$ the translation vector is $R_\beta a + b$.

Problem 5.10

1. $0 \in U$ (since it is a subspace) and so $a \in \mathcal{A}$. Affine subspaces of dimension 0 are also called vectors. In $\mathbb{R}^2$, an affine subspace of dimension 1 is obtained by the parallel translation of a line by vector a (the same is true for $\mathbb{R}^n$). In $\mathbb{R}^3$, an affine subspace of dimension 2 is obtained by the parallel translation of a plane with vector a.

An affine combination of vectors $a + u_i \in \mathcal{A}$ is given by

$$\sum_i \lambda_i(a + u_i) = a + \sum_i \lambda_i u_i$$

since $\sum_i \lambda_i = 1$ and since U is a subspace, $a + \sum_i \lambda_i u_i \in \mathcal{A}$.

2. If $b \in \mathcal{A}$ then $b = a + u'$ for some $u' \in U$. Since

$$a + u = a + u' + u - u' = b + u - u'$$

and $u - u' \in U$ the desired result follows.

3. Since $a \in \mathcal{A}$, the zero vector is a member of U_a which is an important requirement for U_a to be a subspace. Next note that

$$x - a + y - a = (x + y - a) - a, \qquad x, y, a \in \mathcal{A}.$$

$x+y-a$ is an affine combination and so it is a member of $\mathcal{A}$. From the definition of U_a, $x-a+y-a \in U_a$ and so this is closed under vector addition. Regarding scalar multiplication

$$\lambda(x-a) = (\lambda x - \lambda a + a) - a.$$

$\lambda x - \lambda a + a$ is another affine combination of members of $\mathcal{A}$ and so it is also a member. Therefore $\lambda(x-a) \in \mathcal{A}$. Finally use the property $x \in \mathcal{A} \Leftrightarrow x - a \in U_a$ to conclude that $\mathcal{A} = a + U_a$.

If $a, b \in \mathcal{A}$ then assume that it is possible to have $u \in U_a$ with $u \notin U_b$. Since $u = x - a = x + b - a - b$ and $x, a, b \in \mathcal{A}$, $x + b - a$ is an affine combination and so it is also a member of $\mathcal{A}$. Hence $x - a \in U_b$ and we have a contradiction. So $U_a = U_b$; it is also true that $U_a = U$ since if $u \in U_a,\ x,\ z \in \mathcal{A}$

$$u = z - a = z - a + x - x = y - x$$

with $y \in \mathcal{A}$ since $z - a + x$ is an affine combination.

4. If two affine subspaces $\mathcal{A}$, $\mathcal{B}$ have the same direction then $\mathcal{A} = a + U$ and $\mathcal{B} = b + U$ for some $a \in \mathcal{A}$ and $b \in \mathcal{B}$. If $\mathcal{A} \neq \mathcal{B}$, then assume that, for some $u, u' \in U$, $a + u = b + u'$. For any element $v + a \in \mathcal{A}$ we can also write it as $v + u' - u + b$. Since $v, u, u' \in U$, $v + u' - u \in U$ and so $v + a \in \mathcal{B}$. Therefore $\mathcal{A} = \mathcal{B}$ and we have reached a contradiction. We conclude that $\mathcal{A} \cap \mathcal{B} = \varnothing$ if $\mathcal{A} \neq \mathcal{B}$ and they are parallel.

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